
3d Computer Vision

§2 Geometry

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1 Affine Geometry

- In general, assume an
 - orthonormal (mutually perpendicular and normalized basis vectors) or
 - Cartesian (orthonormal and right-handed)coordinate system.

Given:

- Coordinate system S' (e.g., world system) given by
$$S': (O'; e'_1, e'_2) \quad \text{or} \quad S': (O'; e'_1, e'_2, e'_3)$$
- Coordinate system S (e.g., object system) given by
$$S: (O; e_1, e_2) \quad \text{or} \quad S: (O; e_1, e_2, e_3).$$

Wanted:

- Transformation from S to S' or
- Change of a point's coordinates when changing from S to S' .

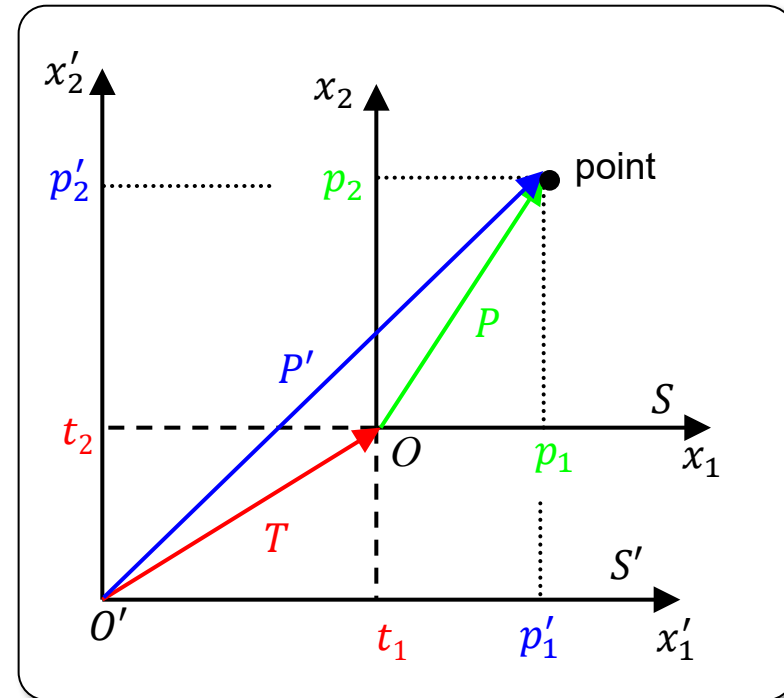
1.1 2d Affine Geometry

Example 1: Translation

Assumption: Coordinate axes are pairwise parallel.

For the system S' this yields:

- S' results from S by a translation by $-T$, where $T = (t_1, t_2)^t$ are the coordinates of the origin O of S in the system S' .
- The point has
 - in S the coordinates $P = (p_1, p_2)^t$,
 - in S' the coordinates $P' = T + P = (t_1, t_2)^t + (p_1, p_2)^t$.



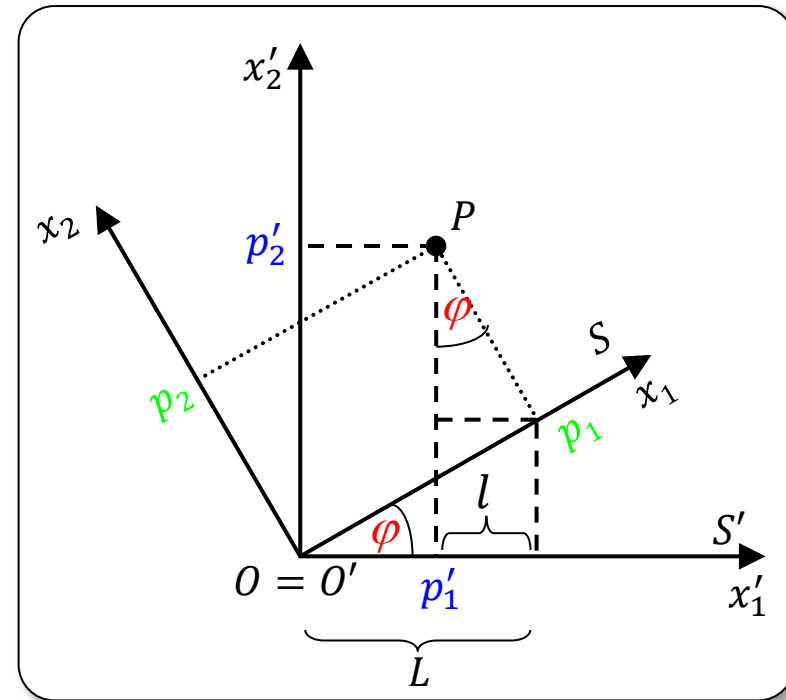
1.1 2d Affine Geometry

Example 2: Rotation

Assumption: Both coordinate systems have the same origin $O = O'$.

Rotation of the system S relative to S' by an angle φ around O :

- The system S' results from S by a rotation by $-\varphi$.

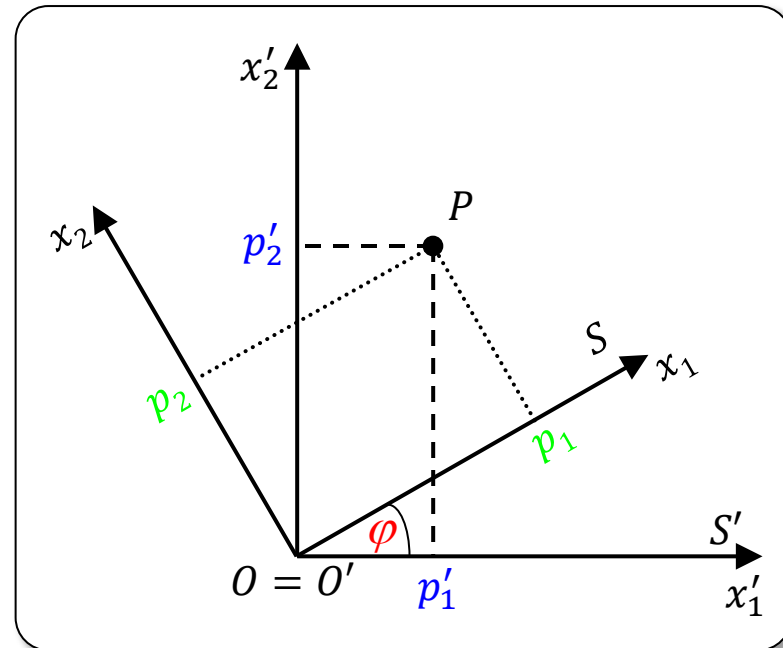


- We get $l/p_2 = \sin \varphi$ and $L/p_1 = \cos \varphi$
thus $p'_1 = L - l = p_1 \cos \varphi - p_2 \sin \varphi$
analog $p'_2 = \dots = p_1 \sin \varphi + p_2 \cos \varphi$

1.1 2d Affine Geometry

- Short form

$$\begin{aligned} P' = \begin{pmatrix} p'_1 \\ p'_2 \end{pmatrix} &= \begin{pmatrix} p_1 \cos \varphi - p_2 \sin \varphi \\ p_1 \sin \varphi + p_2 \cos \varphi \end{pmatrix} \\ &= \begin{pmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} \end{aligned}$$



- In vector-matrix-notation:

$$P' = R(\varphi) \cdot P,$$

with an orthonormal rotation matrix: $R(\varphi) = \begin{pmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{pmatrix}$

- **Remark:** R is orthonormal, iff* $R^{-1} = R^t$.

* iff = if and only if

1.1 2d Affine Geometry

Remark on orthonormal matrices

- A 2×2 matrix R is called **orthonormal**, if its columns r_1, r_2 satisfy the **orthonormality conditions**

$$\left. \begin{array}{l} r_1^t \cdot r_1 = r_2^t \cdot r_2 = 1, \\ r_1^t \cdot r_2 = r_2^t \cdot r_1 = 0, \end{array} \right\} \iff r_i^t \cdot r_j = \delta_{ij}, i, j = 1, 2.$$

■ Properties:

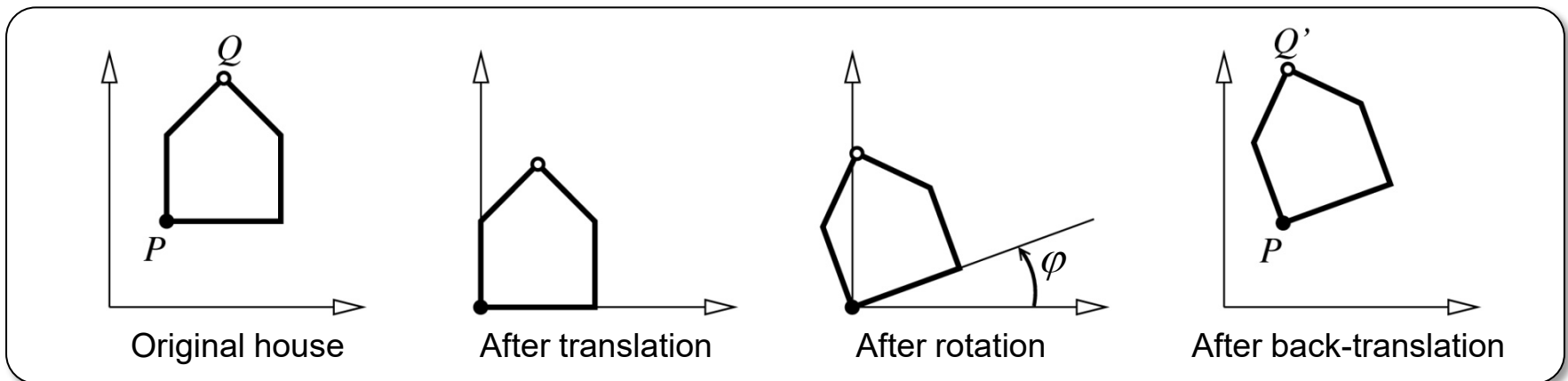
1. A matrix R is orthonormal, iff $R^{-1} = R^t$.
2. The length of mapped vectors does not change.
3. The angles between mapped vectors do not change.
4. One degree of freedom suffices to uniquely define an orthonormal 2×2 matrix with positive determinant (aka orientation)

■ Examples: Rotation, reflections, rotary reflection.

1.1 2d Affine Geometry

Example 3: Rotation around an arbitrary point P

- (1) translation of P into the origin,
- (2) rotation around the origin around the angle φ ,
- (3) translation of P to its original position.



Remark: Matrix multiplication is not commutative.

- The order of matrices must be the same as the order of rotations .

1.1 2d Affine Geometry

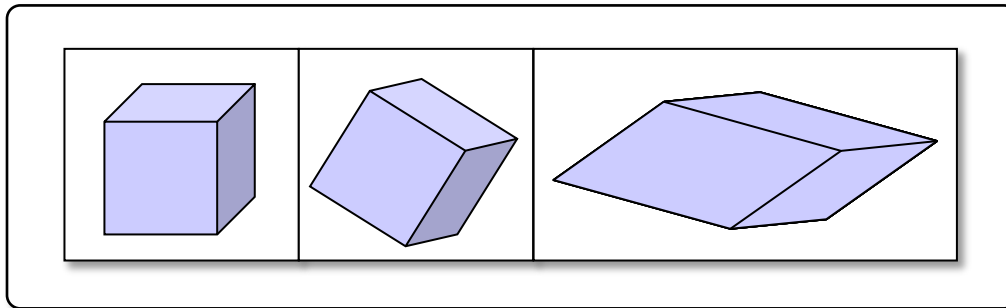
Example 4: Scaling

- To enlarge or shorten the system S , a scaling is used:

$$p'_1 = \lambda_1 \cdot p_1$$

$$p'_2 = \lambda_2 \cdot p_2$$

- In vector-matrix-notation: $P' = S \cdot P$ with $S = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$.
- **Attention!** An arbitrary scaling can change lengths and angles, i.e., orthogonality (angles), orthonormality (length), right-handedness (orientation)!



1.1 2d Affine Geometry

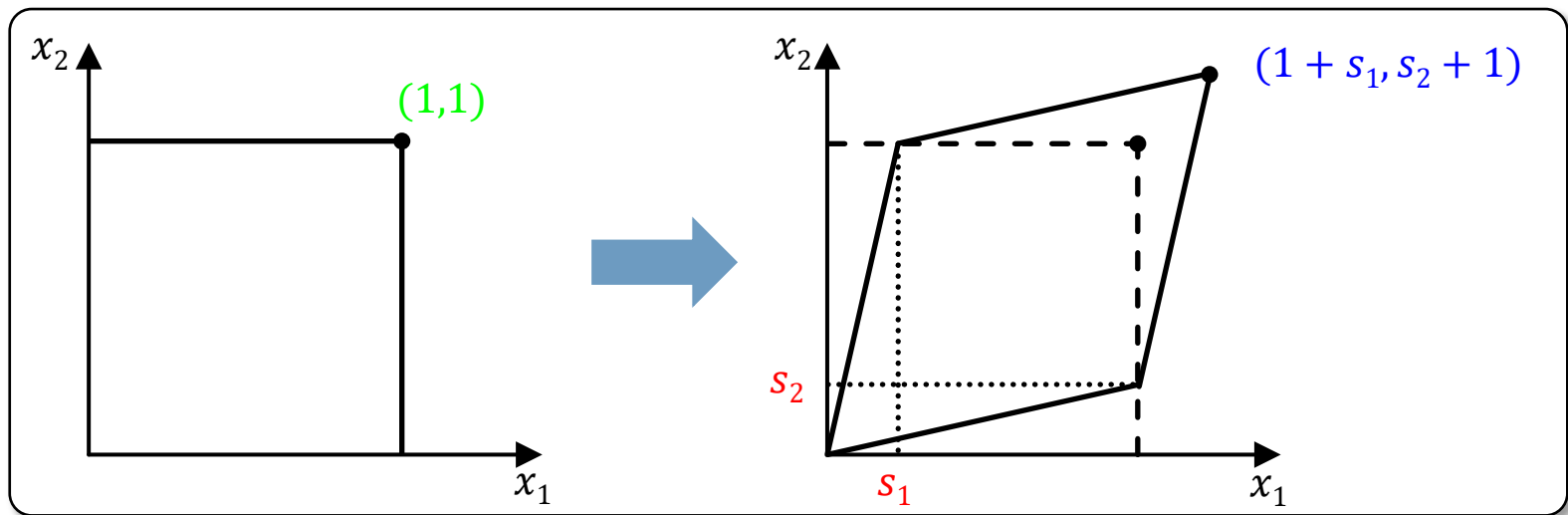
Example 5: Shearing

- A Shearing is given by:

$$p'_1 = p_1 + s_1 \cdot p_2$$

$$p'_2 = s_2 \cdot p_1 + p_2$$

- In Vector-matrix-notation: $P' = S \cdot P$ with $S = \begin{pmatrix} 1 & s_1 \\ s_2 & 1 \end{pmatrix}$.



1.1 2d Affine Geometry

Example 6: Reflections

- Reflection at the line $y = x$: $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.
- Reflection at the line $y = -x$: $A = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$.
- Reflection at the x -axis: $A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.
- Reflection at the y -axis: $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.
- Reflection at the origin: $A = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$.

1.1 2d Affine Geometry

Definition (2d)

- An **affine map** (aka affine transformation) is a combination of a linear map $A \in \mathbb{R}^{2 \times 2}$ and a translation $T \in \mathbb{R}^2$:

$$P' = A \cdot P + T \text{ with } A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \text{ and } T = \begin{pmatrix} a_{10} \\ a_{20} \end{pmatrix}.$$

- Translations, rotations, scalings, shearings, reflections and combinations of these are affine transformations.

1.1 2d Affine Geometry

Properties

1. Ratios remain unchanged (aka **affine invariance**), i.e., for points P and Q application of an affine map F yields

$$F(\lambda P + (1-\lambda)Q) = \lambda F(P) + (1-\lambda)F(Q) \quad \text{for } 0 \leq \lambda \leq 1$$

and for general points P_i

$$F\left(\sum_{i=0}^n \alpha_i P_i\right) = \sum_{i=0}^n \alpha_i F(P_i), \quad \sum_{i=0}^n \alpha_i = 1$$

2. Parallel lines remain parallel.
 3. Angles, lengths, directions and orientations can change.
 4. To determine its six parameters a_{ij} , three (non-collinear) corresponding points in both coordinate systems are necessary.
- In general, a square is mapped to a rotated and translated parallelogram.

1.1 2d Affine Geometry

Definition (2d)

- A **similarity transformation** is a combination of a uniform scaling $s \in \mathbb{R}$, a rotation $R \in SO(2) \subseteq \mathbb{R}^{2 \times 2}$, and a translation $T \in \mathbb{R}^2$:

$$P' = s \cdot R \cdot P + T \text{ with } R = \begin{pmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{pmatrix} \text{ and } T = \begin{pmatrix} r_{10} \\ r_{20} \end{pmatrix}.$$

- **Properties**

1. Ratios, angles, and parallel lines are preserved.
 2. Length, directions and orientations can change.
 3. To determine its four parameters $r_{10}, r_{20}, s, \alpha$ at least two corresponding points in both coordinate systems are necessary.
- ▶ A square is mapped to a scaled, rotated, and translated square.

1.1 2d Affine Geometry

Definition (2d)

- A **rigid body motion** is a combination of an orthogonal transformation $A \in O(2) \subseteq \mathbb{R}^{2 \times 2}$ and a translation $T \in \mathbb{R}^2$:

$$P' = A \cdot P + T \text{ with } R = \begin{pmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{pmatrix} \text{ and } T = \begin{pmatrix} r_{10} \\ r_{20} \end{pmatrix}.$$

- **Properties**

1. It is a combination of a rotation, translation, and reflection (if $\det A = -1$).
2. Ratios, angles, lengths, and parallel lines are preserved.
3. Orientation can change, if $\det A = -1$ (aka **improper rigid motion**).
 - If $\det A = 1$ the map is a so-called **proper rigid motion**.
 - ▶ A square is mapped to a rotated and translated square.

1.1 2d Affine Geometry

Homogeneous coordinates

- The serial application of a rotation R , translation T , and scaling S yields the equation

$$P' = S \cdot (T + R \cdot P).$$

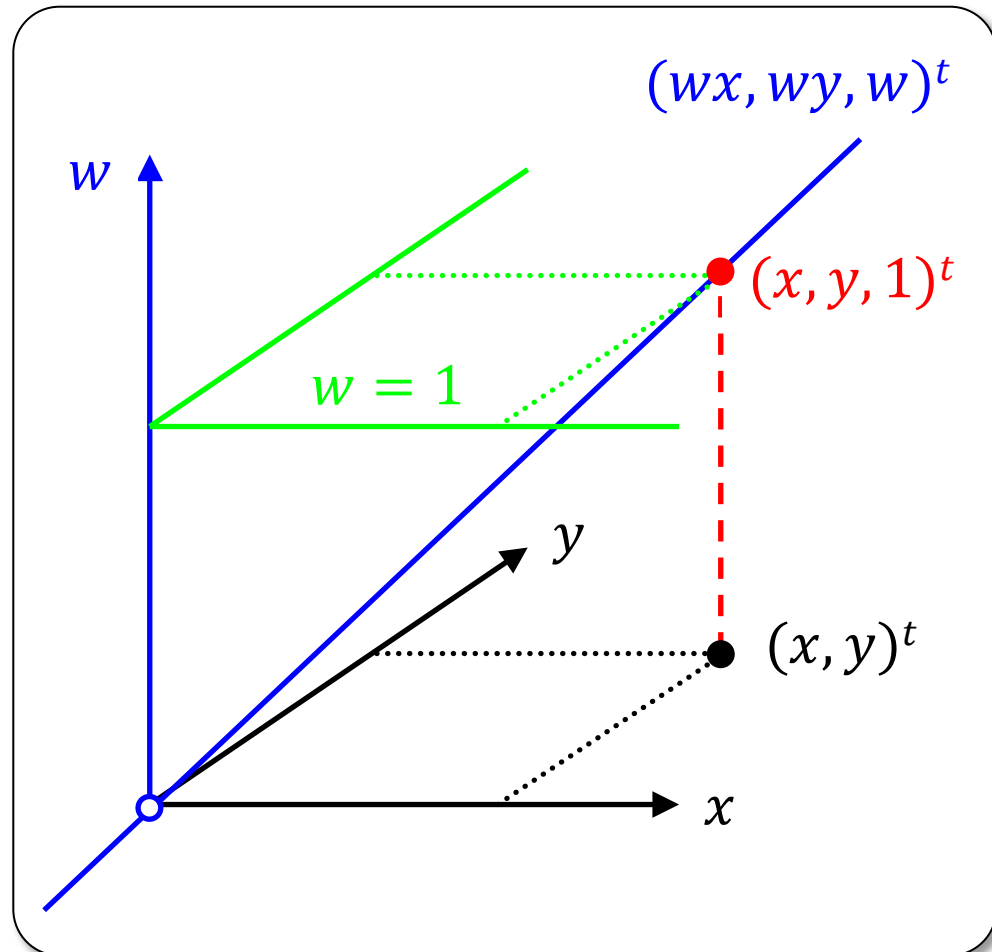
- If multiple of these transformation must be computed, the addition in the above equation is annoying.
- Today's GPUs implement matrix multiplications in hardware.
- It would be much better, if all affine transformations were represented as matrix multiplications

$$P' = M_n \cdots M_3 \cdot M_2 \cdot M_1 \cdot P.$$

1.1 2d Affine Geometry

Lifting the problem to the next higher dimension:

- ▶ The triple $(wx, wy, w)^t$, $w \neq 0$, represents the homogenous coordinates of the point $(x, y)^t \in \mathbb{R}^2$.
- ▶ There are infinitely many such representations for the same point.
- ▶ Use the so-called standard representation $w = 1$.
- ▶ Now, a point $P = (x, y)^t \in \mathbb{R}^2$ has the unique homogenous coordinates $(x, y, 1)^t$.
- Analogously for points in $\mathbb{R}^3$.



1.1 2d Affine Geometry

Representation of affine transformations in homogenous coordinates:

- **Translation** of point $(x, y)^t$ by vector $(t_1, t_2)^t$:

$$\begin{pmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} x + t_1 \\ y + t_2 \\ 1 \end{pmatrix} = \begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix}$$

- **Rotation** of point $(x, y)^t$ around angle φ around the origin:

$$\begin{pmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} x \cdot \cos \varphi - y \cdot \sin \varphi \\ x \cdot \sin \varphi + y \cdot \cos \varphi \\ 1 \end{pmatrix} = \begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix}$$

1.1 2d Affine Geometry

- **Scaling** of point $(x, y)^t$ by factors λ_1 and λ_2 :

$$\begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} \lambda_1 \cdot x \\ \lambda_2 \cdot y \\ 1 \end{pmatrix} = \begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix}.$$

- **Rotation** of point $(x, y)^t$ **around a point** $P = (p_x, p_y)^t$ around angle φ :

$$\begin{pmatrix} 1 & 0 & p_x \\ 0 & 1 & p_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -p_x \\ 0 & 1 & -p_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix}$$

1.2 3d Affine Geometry

Definitions (3d)

- An **affine map** is a combination of a linear map $A \in \mathbb{R}^{3 \times 3}$ and a translation $T \in \mathbb{R}^3$

$$P' = A \cdot P + T.$$

- A **similarity transformation** is a combination of a rotation $R \in \mathbb{R}^{3 \times 3}$, a scaling $s \in \mathbb{R}$ and a translation $T \in \mathbb{R}^3$

$$P' = s \cdot R \cdot P + T.$$

- **Properties**

1. Ratios, parallel lines, angles, lengths, and orientations change analog to the planar cases.
2. To determine its 12 resp. 7 parameters, four resp. three corresponding points in both coordinate systems are necessary.

1.2 3d Affine Geometry

Definitions (3d)

- A **rigid body motion** is a combination of an orthogonal transformation $R \in O(3) \subseteq \mathbb{R}^{3 \times 3}$ and a translation $T \in \mathbb{R}^3$

$$P' = R \cdot P + T.$$

- **Properties**

1. It is a combination of a rotation, translation, and reflection (if $\det A = -1$).
2. Ratios, angles, lengths, and parallel lines are preserved.
3. Orientations can change, if $\det A = -1$ (aka **improper rigid motion**).
 - If $\det A = 1$ the map is a so-called **proper rigid motion**.

1.2 3d Affine Geometry

- **Translation:** The translation of a point $(x, y, z)^t$ along the translation vector $(t_x, t_y, t_z)^t$ yields the point $(x', y', z')^t$ with

$$\underbrace{\begin{pmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix}}_{=T(t_x, t_y, t_z)} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x + t_x \\ y + t_y \\ z + t_z \\ 1 \end{pmatrix} = \begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix}.$$

- **Scaling:** A scaling by the factors λ_1, λ_2 , and λ_3 in all three axis directions yields

$$\begin{pmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 \\ 0 & 0 & \lambda_3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} \lambda_1 \cdot x \\ \lambda_2 \cdot y \\ \lambda_3 \cdot z \\ 1 \end{pmatrix} = \begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix}.$$

1.2 3d Affine Geometry

Question

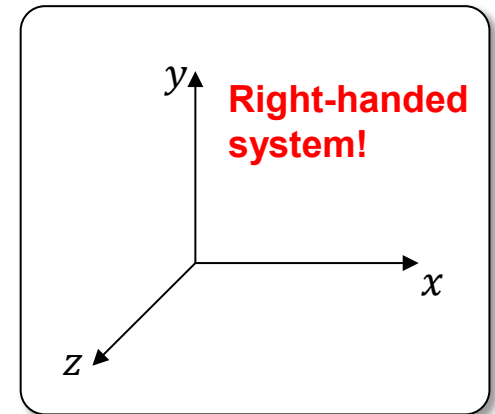
- How many degrees of freedom does a general 3d-rotation have, i.e.,
 - arbitrary axis of rotation and
 - arbitrary angle of rotation?

1.2 3d Affine Geometry

Rotation

- All rotations in mathematically positive orientation (i.e., counter-clockwise).
 - The observer sits on the axis of rotations and looks towards the origin of the coordinate system.
- First only rotations around a single coordinate axis around the angle φ .
- ▶ Transformation matrices $R_x(\varphi), R_y(\varphi), R_z(\varphi)$.
- We use the interpretation:

„global fixed coordinate system;
point is transformed (rotated)“

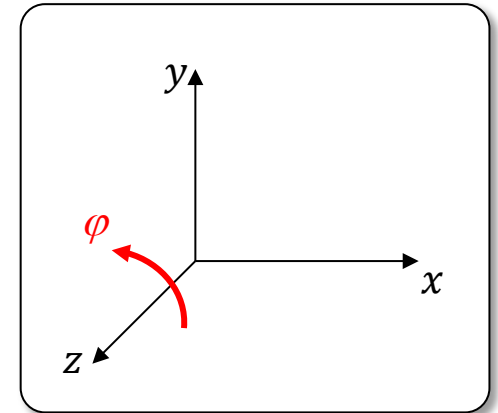


1.2 3d Affine Geometry

Rotation around the z-axis

- The rotation of a point $(x, y, z)^t$ around the z-axis around the angle φ yields point $(x', y', z')^t$ with

$$\underbrace{\begin{pmatrix} \cos \varphi & -\sin \varphi & 0 & 0 \\ \sin \varphi & \cos \varphi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}_{=R_Z(\varphi)} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x \cdot \cos \varphi - y \cdot \sin \varphi \\ x \cdot \sin \varphi + y \cdot \cos \varphi \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix}$$



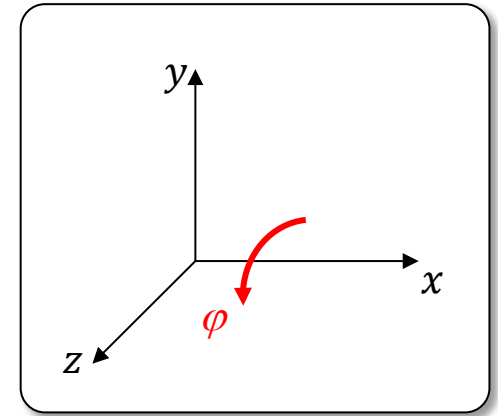
- **Note:** Rotation around the angle φ around the z-axis corresponds to the 2d-case, where the z-coordinate is constant!

1.2 3d Affine Geometry

Rotation around the x -axis

- The rotation of a point $(x, y, z)^t$ around the x -axis around the angle φ yields point $(x', y', z')^t$ with

$$\underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \varphi & -\sin \varphi & 0 \\ 0 & \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}_{=R_x(\varphi)} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ y \cdot \cos \varphi - z \cdot \sin \varphi \\ y \cdot \sin \varphi + z \cdot \cos \varphi \\ 1 \end{pmatrix} = \begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix}$$



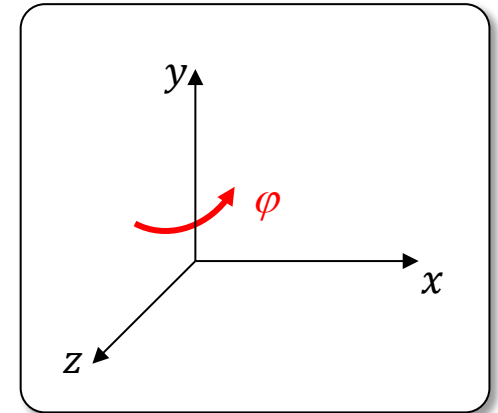
- **Note:** Rotation around the angle φ around the x -axis corresponds to the 2d-case, where the x -coordinate is constant!

1.2 3d Affine Geometry

Rotation around the y -axis

- The rotation of a point $(x, y, z)^t$ around the y -axis around the angle φ yields point $(x', y', z')^t$ with

$$\underbrace{\begin{pmatrix} \cos \varphi & 0 & \sin \varphi & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \varphi & 0 & \cos \varphi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}_{=R_y(\varphi)} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x \cdot \cos \varphi + z \cdot \sin \varphi \\ y \\ -x \cdot \sin \varphi + z \cdot \cos \varphi \\ 1 \end{pmatrix} = \begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix}$$



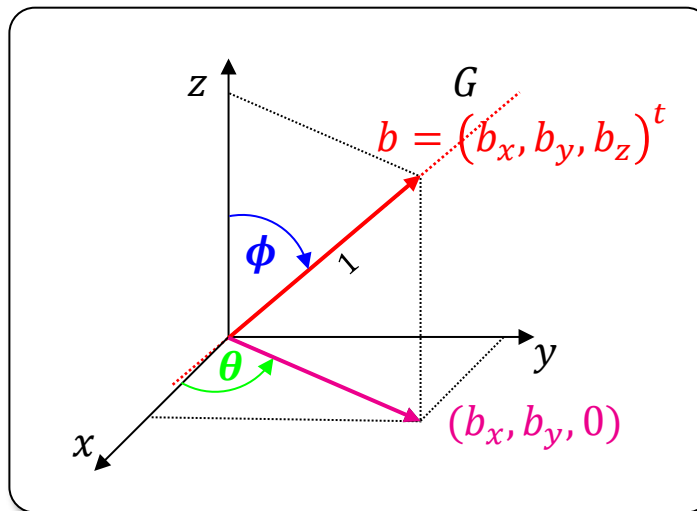
- **Note:** Rotation around the angle φ around the y -axis corresponds to the 2d-case, where the y -coordinate is constant!

1.2 3d Affine Geometry

Rotation around arbitrary axis G

- **Goal:** Rotation $R_G(\psi)$ of a point P around an arbitrarily aligned axis G in space around an angle ψ .
- **Special case:** Axis of rotation G passes through the origin and is spanned by vector $\mathbf{b} = (b_x, b_y, b_z)^t$ with $\|\mathbf{b}\| = 1$:

$$G = \{ \lambda \cdot \mathbf{b} : \lambda \in \mathbb{R} \}.$$



$$\Rightarrow \begin{cases} b_x = \sin \phi \cdot \cos \theta \\ b_y = \sin \phi \cdot \sin \theta \\ b_z = \cos \phi \end{cases}$$

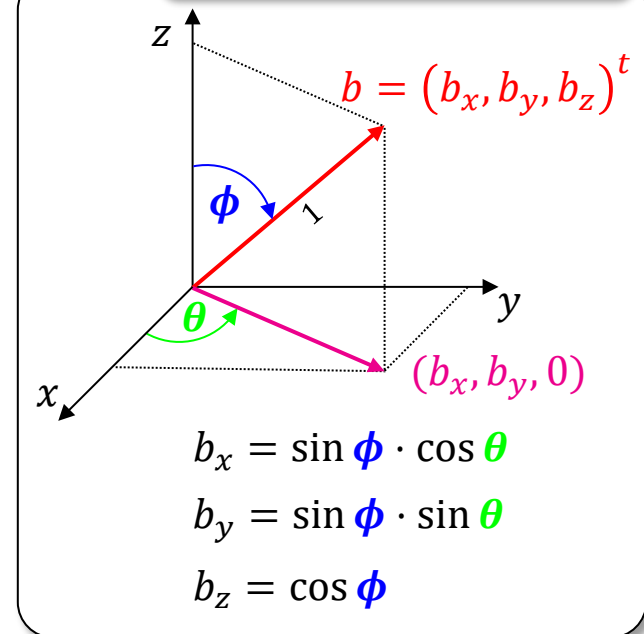
Rem.: Always right-handed system, here only rotated.

1.2 3d Affine Geometry

Rotation around arbitrary axis G

- Every rotation around an arbitrary axis can be combined from **three** rotations around individual coordinates axis.
 - *Intrinsic representation*: angles with respect to the rotated system.
 - *Extrinsic representation*: angles with respect to a fixed world system.
- Choice of axis:
 - *Proper Euler angles*: x-y-x, x-z-x, y-z-y, y-x-y, z-x-z, **z-y-z**
 - *Tait-Bryan angles* (Cardan angles): x-y-z, x-z-y, y-x-z, y-z-x, z-x-y, z-y-x
- ▶ There are 24 different representations in terms of three angles, which are all equivalent.
- ▶ We will use in the sequel the extrinsic z-y-z representation.

Rem.: Always right-handed system, here only rotated.



1.2 3d Affine Geometry

Wanted: Coordinates of a point P after a rotation around the axis G around the angle ψ .

- Extrinsic z-y-z approach:

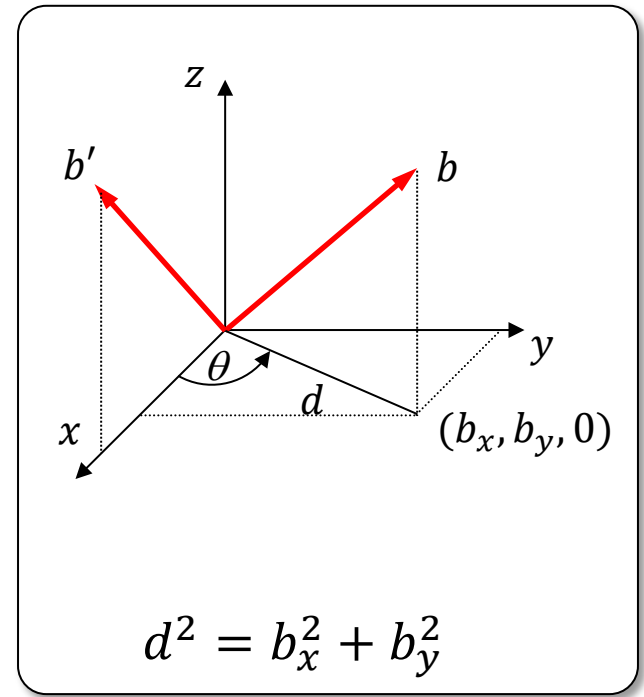
- (1) Transform point P such that the axis of rotation coincides with the z -axis.
 - (2) Then, use the rotation matrix $R_z(\psi)$ for the rotation around ψ .
 - (3) Finally, invert the auxiliary transformation from step (1).
- If G is already the z -axis, the auxiliary transformations are obsolete.

1.2 3d Affine Geometry

Step 1:

- Rotate vector b into the (z, x) -plane: $b \rightarrow b'$.
- P is mapped to P' with $P' = R_z(-\theta) \cdot P$, with

$$R_z(-\theta) = \begin{pmatrix} \cos \theta & \sin \theta & 0 & 0 \\ -\sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$
$$= \frac{1}{d} \begin{pmatrix} b_x & b_y & 0 & 0 \\ -b_y & b_x & 0 & 0 \\ 0 & 0 & d & 0 \\ 0 & 0 & 0 & d \end{pmatrix}.$$

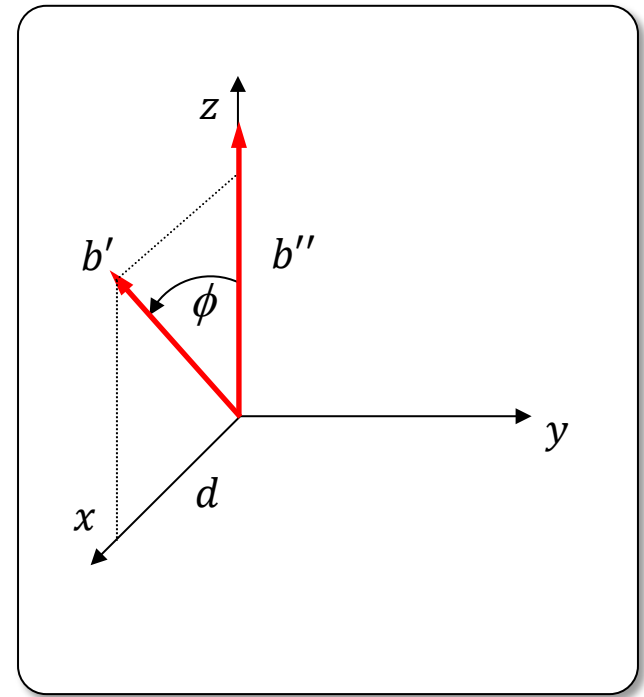


1.2 3d Affine Geometry

Step 2:

- Rotate vector b' onto the z -axis: $b' \rightarrow b''$.
- P' is mapped to P'' with $P'' = R_y(-\phi) \cdot P'$, with

$$R_y(-\phi) = \begin{pmatrix} \cos \phi & 0 & -\sin \phi & 0 \\ 0 & 1 & 0 & 0 \\ \sin \phi & 0 & \cos \phi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$
$$= \begin{pmatrix} b_z & 0 & -d & 0 \\ 0 & 1 & 0 & 0 \\ d & 0 & b_z & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

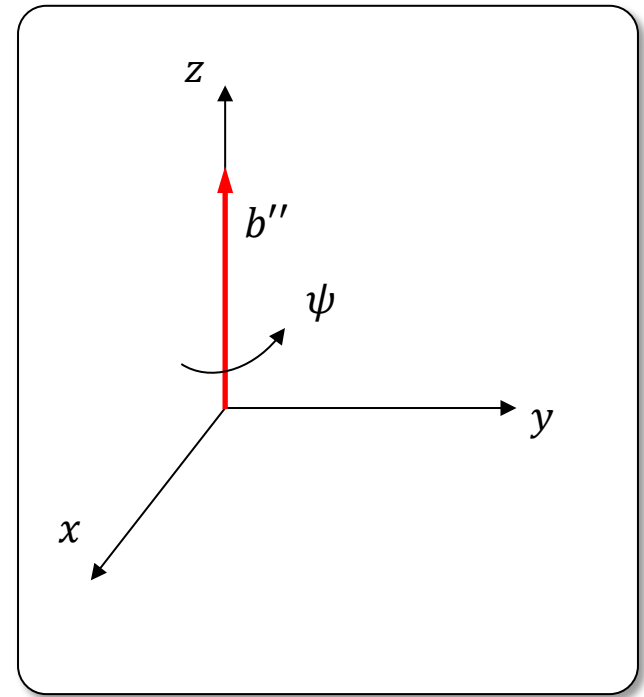


1.2 3d Affine Geometry

Step 3:

- Rotate around the angle ψ around the z-axis.
- P'' mapped to P''' with $P''' = R_z(\psi) \cdot P''$, with

$$R_z(\psi) = \begin{pmatrix} \cos \psi & -\sin \psi & 0 & 0 \\ \sin \psi & \cos \psi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$



1.2 3d Affine Geometry

Steps 4 and 5:

- Invert the rotations from steps 1 and 2 in reverse order.
- P''' is mapped the wanted rotated point Q of the original point P by $Q = R_z(\theta)R_y(\phi) \cdot P'''$, with

$$R_y(\phi) = \begin{pmatrix} b_z & 0 & d & 0 \\ 0 & 1 & 0 & 0 \\ -d & 0 & b_z & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \text{and}$$

$$R_z(\theta) = \frac{1}{d} \begin{pmatrix} b_x & -b_y & 0 & 0 \\ b_y & b_x & 0 & 0 \\ 0 & 0 & d & 0 \\ 0 & 0 & 0 & d \end{pmatrix}.$$

1.2 3d Affine Geometry

Result:

- The total transformation can be realized by the concatenation (multiplication) of the respective individual transformations

$$M_b(\psi) = R_z(\theta)R_y(\phi)R_z(\psi)R_y(-\phi)R_z(-\theta).$$

General case:

If the axis of rotation is an arbitrary line

$$G = \{a + \lambda \cdot b : \lambda \in \mathbb{R}, a = (a_x, a_y, a_z)^t, \|b\| = 1\},$$

before the first step 1 and after the last step 5 a translation must be applied, i.e.

$$M_b(\psi) = T(a) R_z(\theta) R_y(\phi) R_z(\psi) R_y(-\phi) R_z(-\theta) T(-a).$$

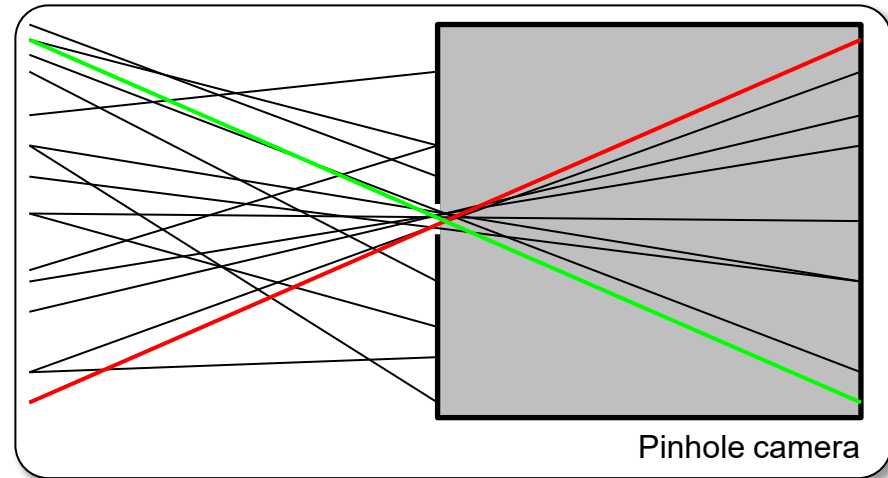
Content

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 - 1.1 2d Affine Geometry ✓
 - 1.2 3d Affine Geometry ✓
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 - 2.1 Parallel Projection
 - 2.2 Perspective Projection
- 3. Physics of Camera Projections*

2. Projective Geometry

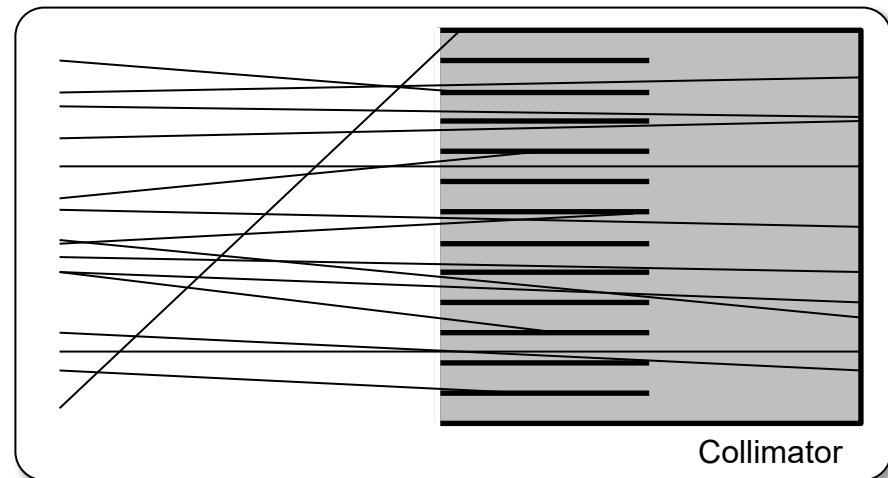
Pinhole camera: All rays pass through one hole.

- The hole is ideally a single point, the **center of projection** N .
- ▶ (Almost) radial rays from N at the image plane.
- ▶ The image is upside down.



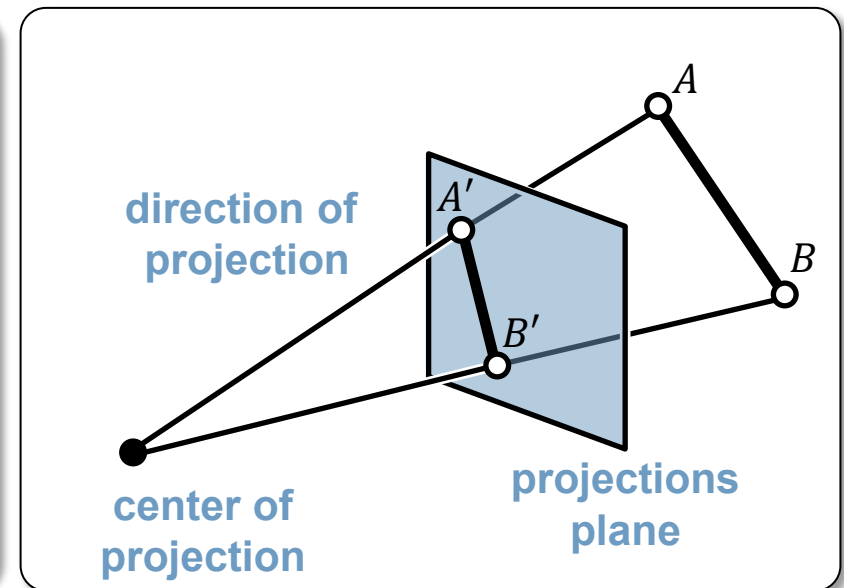
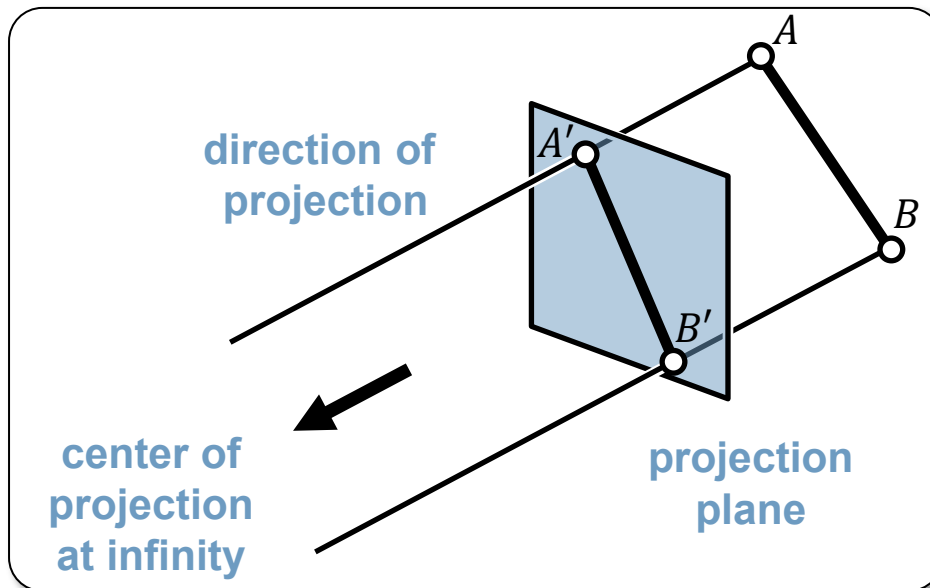
Collimator: All rays pass through a system of parallel tubes.

- ▶ (Almost) parallel rays at the image plane.
- ▶ Yields almost parallel rays from a divergent source.



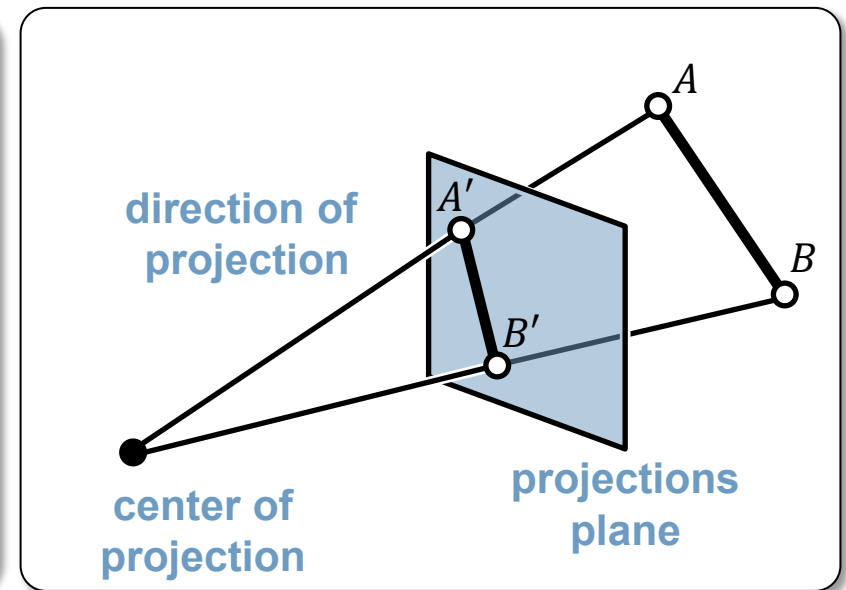
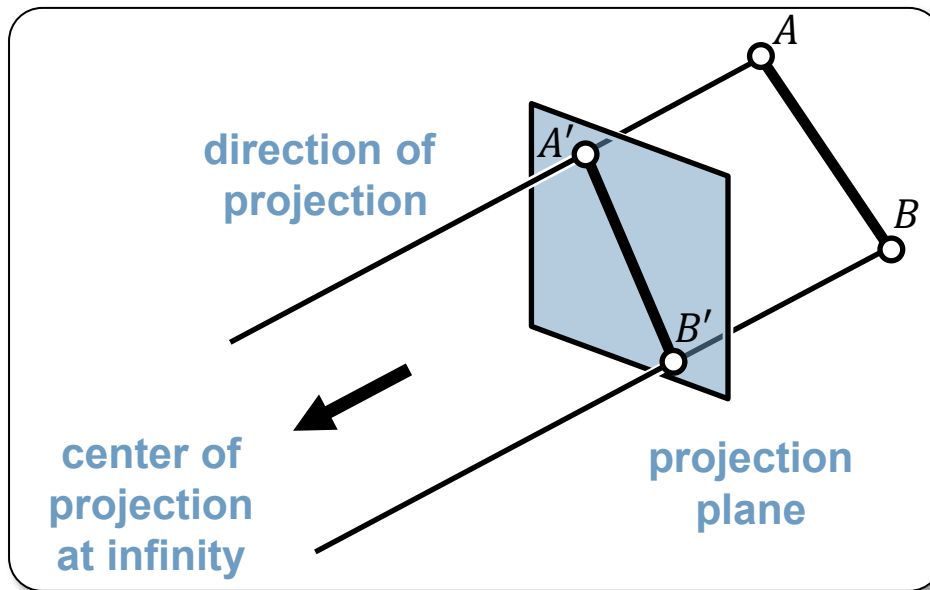
2. Projective Geometry

- **Definition:** A **projection** is a mapping, that maps a space of dimension n to a space of dimension $< n$.
- Because displays and cameras are 2d devices, 3d-object are plotted resp. captured in 2d-views.
 - ▶ A space point A is mapped along a **projection ray** (direction of projection) to a given **projection plane** (image plane).



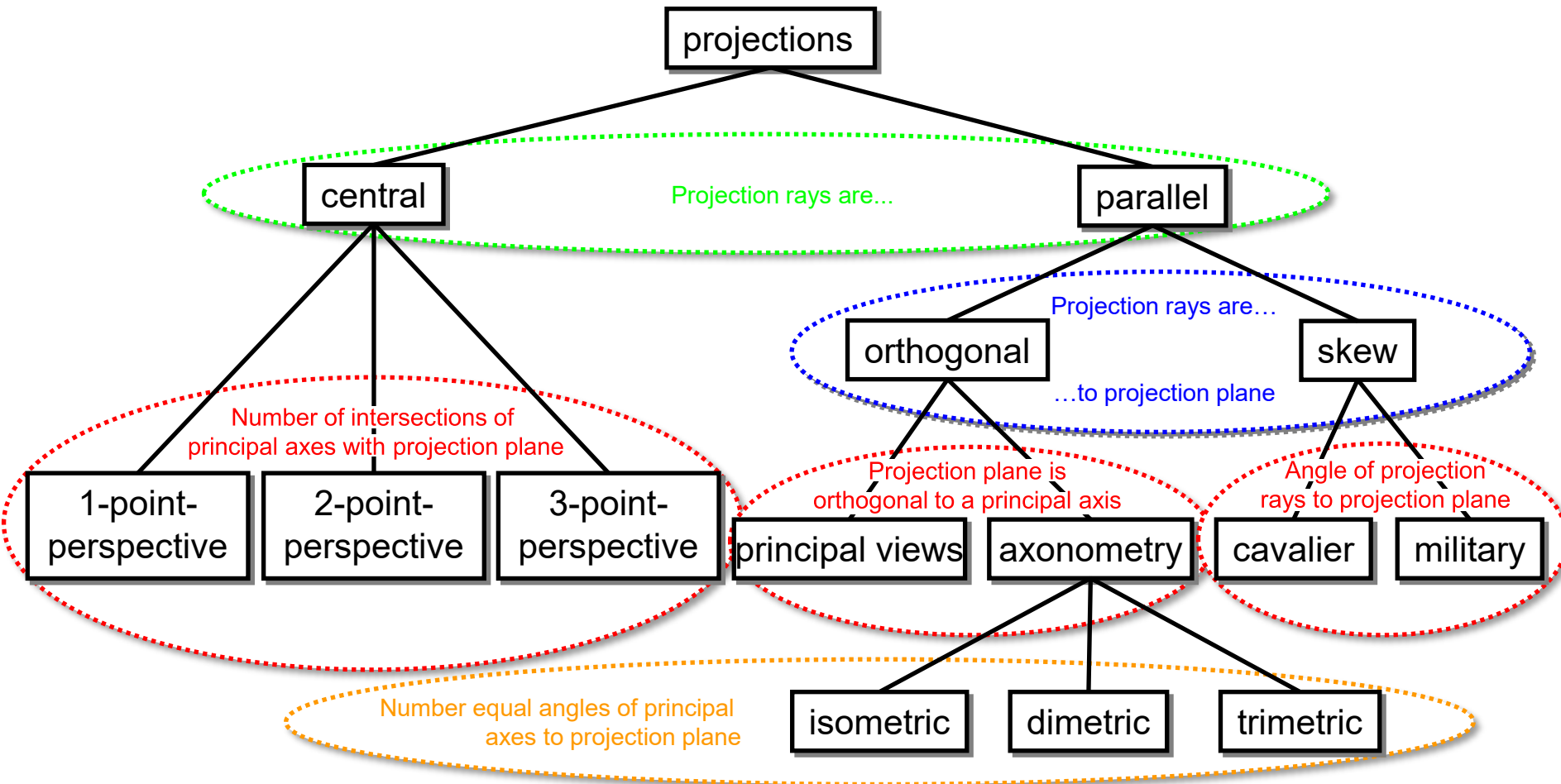
2. Projective Geometry

- The direction of a projection is determined by the **center of projection** and a space point A .
- The intersection of the projection ray with the projection plane determines the **projected space point** A' .



2. Projective Geometry

Classification of projections



2. Projective Geometry

Classification of projections

- Abbreviations: projection ray = p-ray,
projection plane = p-plane,
principal axes = p-axes
- Direction of p-ray:
 - (1) central
 - ▶ relative pose of p-axes to p-plane
 - number of intersections of p-axes with p-plane.
 - (2) parallel
 - ▶ angle of p-ray to p-plane
 - a) orthogonal
 - b) skew
 - ▶ relative pose of p-axes/-ray to p-plane
 - angle of p-axes/-ray to p-plane.

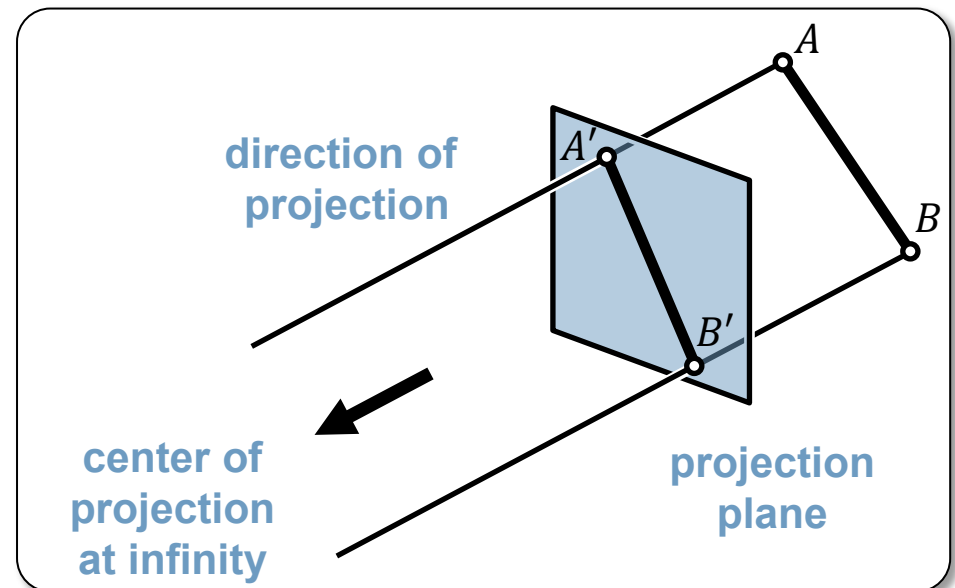
Content

- 1. Affine Geometry ✓
 - 1.1 2d Affine Geometry ✓
 - 1.2 3d Affine Geometry ✓
- 2. Projective Geometry ✓
 - 2.1 Parallel Projection
 - 2.2 Perspective Projection
- 3. Physics of Camera Projections*

2.1 Parallel Projection

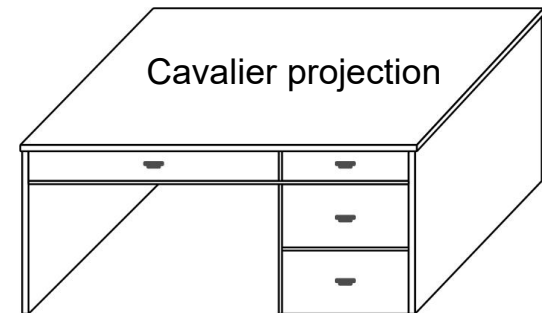
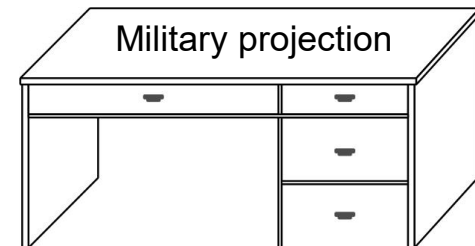
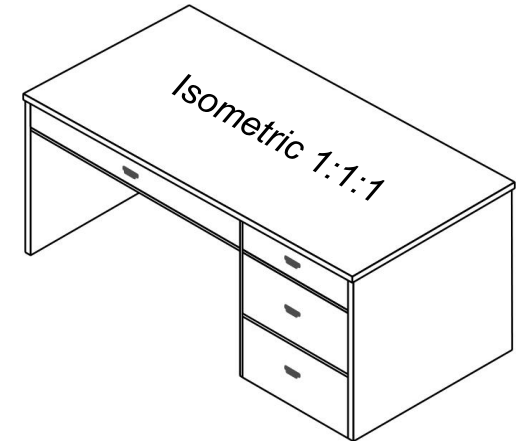
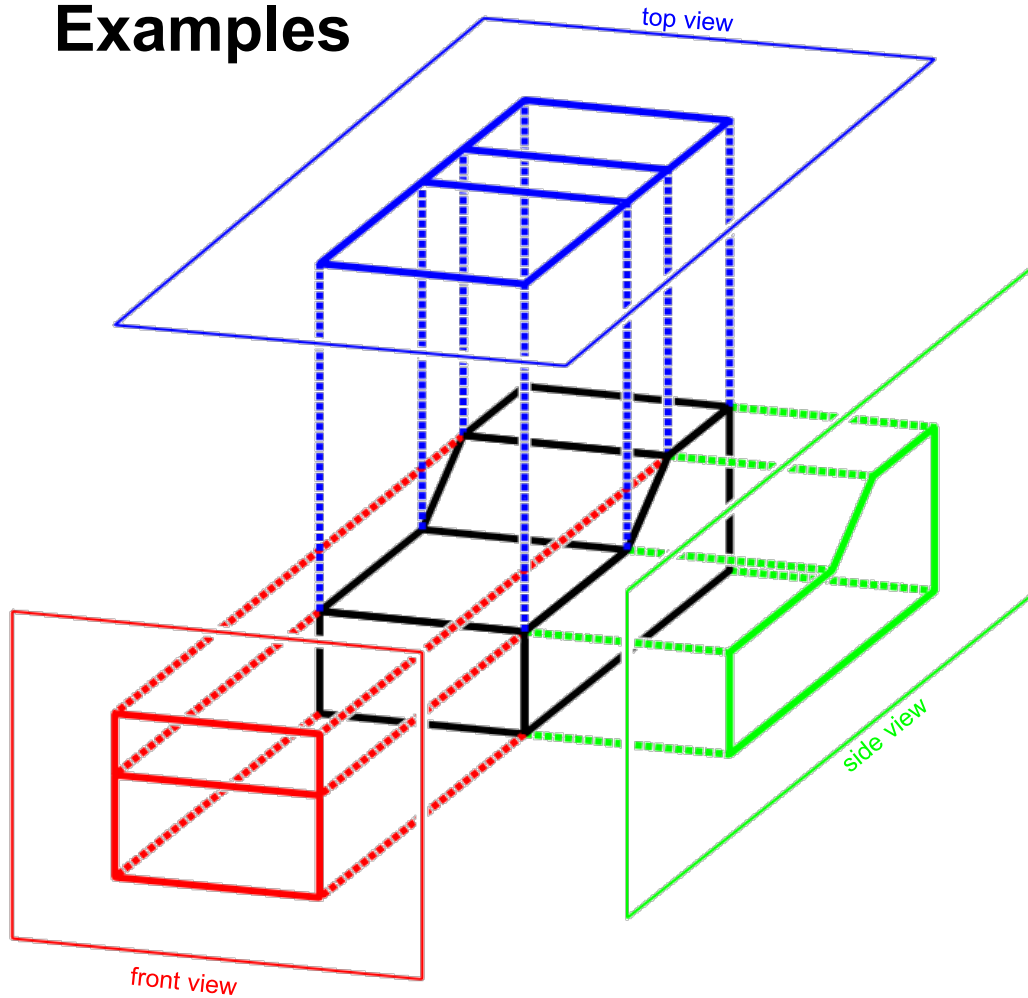
Parallel projection

- All **projection rays** are parallel to one direction.
- The **center of projections** is a point at infinity (dt. Fernpunkt).
- **Disadvantage:** Less realistic.
- **Advantage:** Allows for exact measurements from the image.



2.1 Parallel Projection

Examples



Content

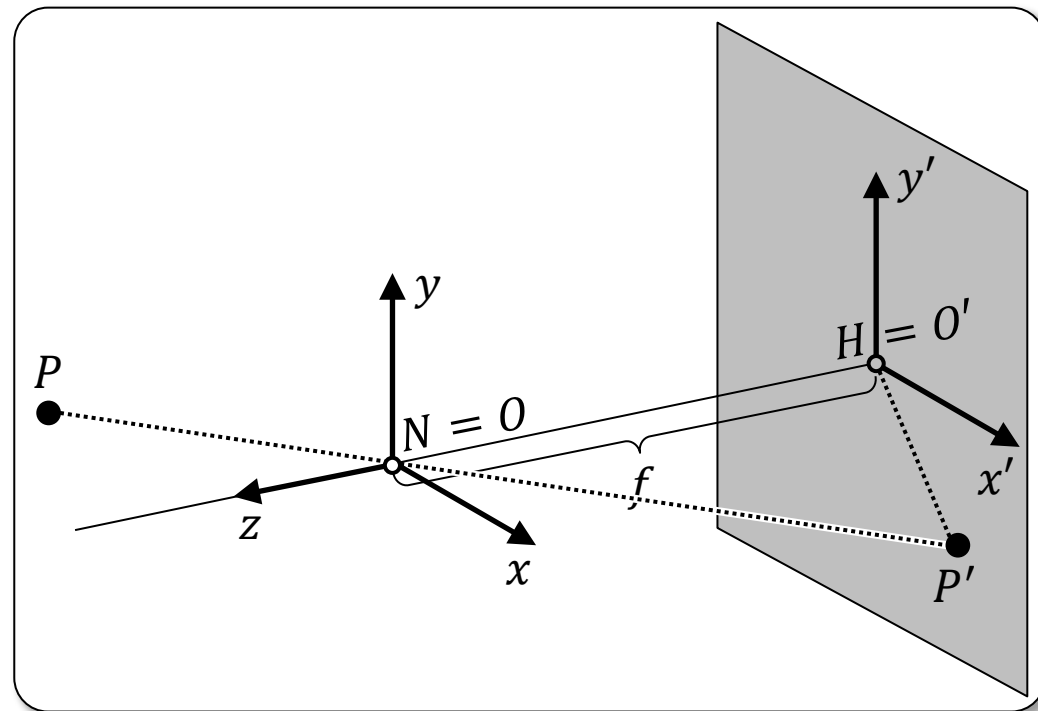
- 1. Affine Geometry ✓
 - 1.1 2d Affine Geometry ✓
 - 1.2 3d Affine Geometry ✓
- 2. Projective Geometry ✓
 - 2.1 Parallel Projection ✓
 - 2.2 Perspective Projection
- 3. Optics of Camera Projections*

2.2 Perspective Projection

Perspective projection (aka central projection)

- **Camera system** $(O; x, y, z)$ in world coordinate system with center N .
- **Image space** $(O'; x', y')$ and **image principal point H** at **principal distance f** , i.e. point on image plane closest to center of projection N .
- **Simplifications:** (1) $N = O$, (2) $H = O'$, and (3) the camera system is the world system with the negative z -axis as view direction.
- **Mapping:**

$$\mathcal{P}: \underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{=P} \in \mathbb{R}^3 \mapsto \underbrace{\begin{pmatrix} x' \\ y' \end{pmatrix}}_{=P'} \in \mathbb{R}^2$$



2.2 Perspective Projection

Perspective projection (aka central projection)

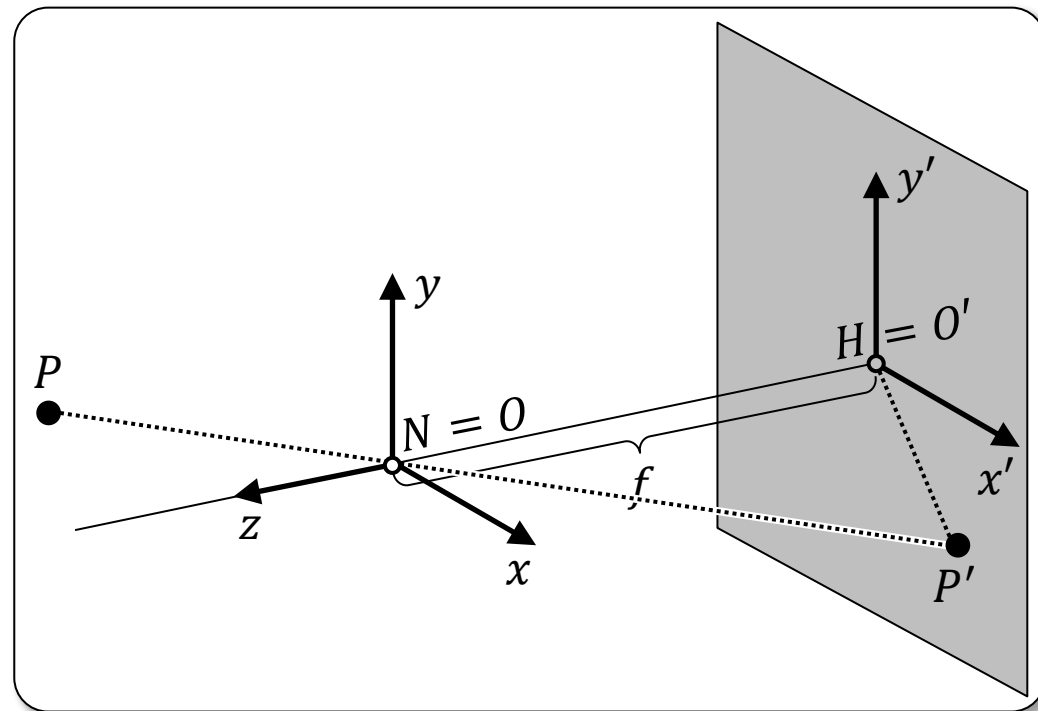
- Mapping (in affine coordinates):

$$\mathcal{P}: \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mapsto \begin{pmatrix} x' \\ y' \end{pmatrix} \in \mathbb{R}^2$$

$$\text{with } \begin{pmatrix} x' \\ y' \end{pmatrix} = -\frac{f}{z} \begin{pmatrix} x \\ y \end{pmatrix}.$$

- In homogeneous coordinates

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} \sim \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$



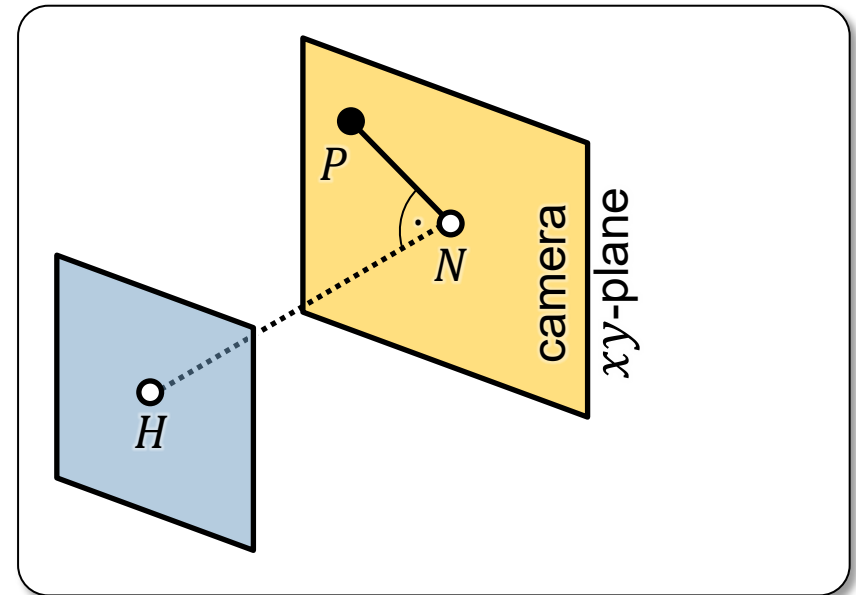
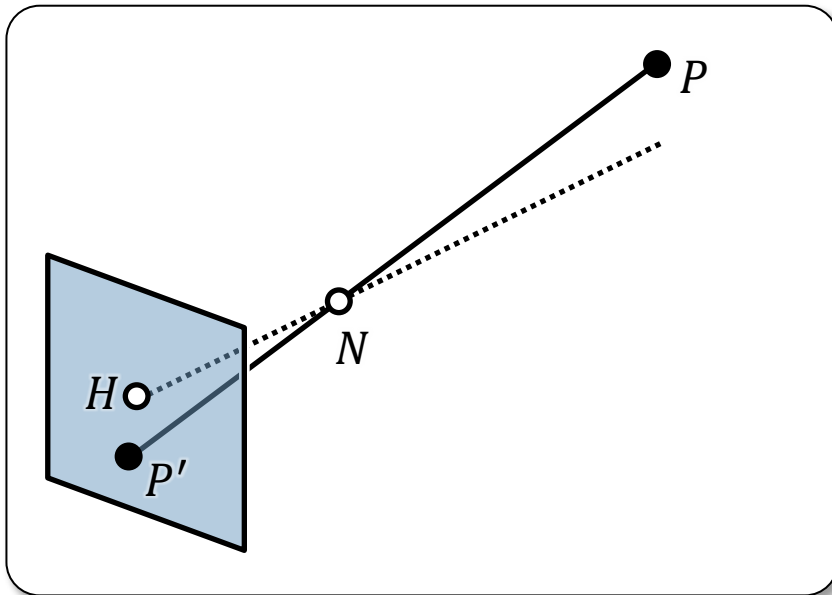
2.2 Perspective Projection

Reminder

$$P' = \begin{pmatrix} x' \\ y' \end{pmatrix} = -\frac{f}{z} \begin{pmatrix} x \\ y \end{pmatrix}$$

Perspective projection of points

- Every 3d-point P is mapped to a 2d-point P' in the **image plane**?
- Only if its z -coordinate is not zero, the point can be projected!
- ▶ If point P is in the **xy -plane of the camera system**, it cannot be projected, because the projection ray is parallel to the image plane.



2.2 Perspective Projection

Reminder

$$P' = \begin{pmatrix} x' \\ y' \end{pmatrix} = -\frac{f}{z} \begin{pmatrix} x \\ y \end{pmatrix}$$

Perspective projection of lines (1)

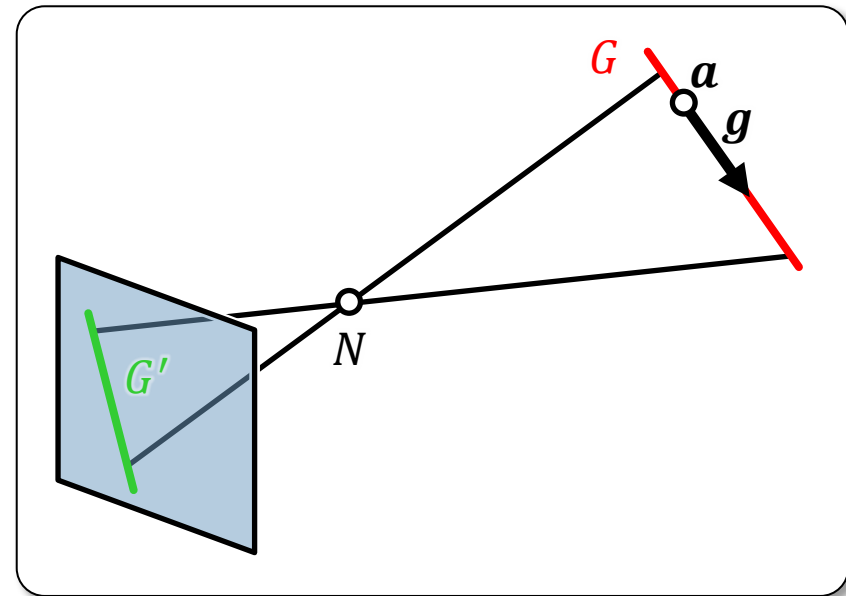
- A 3d-line G is mapped to a 2d-line G' in the **image plane**?
- An affine 3d-line in camera coordinates is a set of points

$$G(\lambda) = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \lambda \begin{pmatrix} g_1 \\ g_2 \\ g_3 \end{pmatrix} = \mathbf{a} + \lambda \mathbf{g}$$

with $\lambda \in \mathbb{R}$.

It is mapped to the affine 2d-line

$$G'(\lambda) = -\frac{f}{a_3 + \lambda g_3} \left(\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \lambda \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} \right).$$



2.2 Perspective Projection

Perspective projection of lines (2)

1. $g_3 = 0, a_3 = 0$: The line is parallel to the **image plane** and lies in the **camera's xy -plane**.

► It cannot be projected.

2. ...

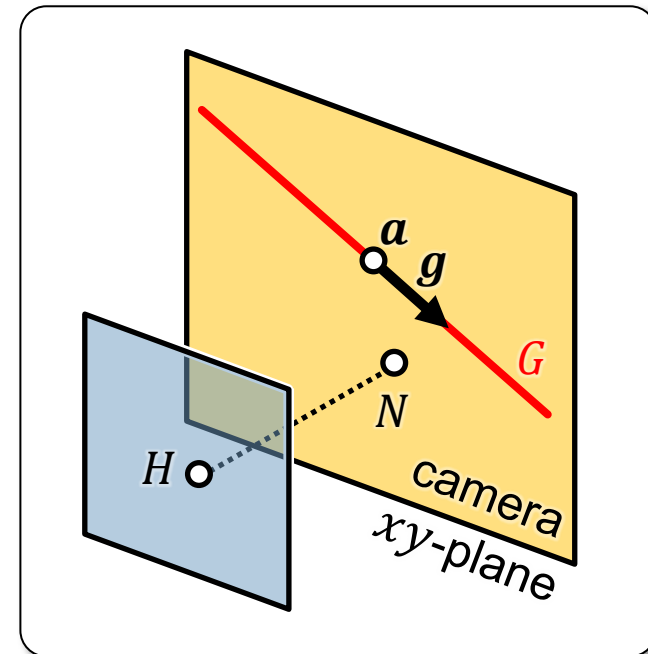
3. ...

4. ...

5. ...

Reminder

$$G'(\lambda) = -\frac{f}{a_3 + \lambda g_3} \left(\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \lambda \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} \right)$$



2.2 Perspective Projection

Perspective projection of lines (3)

1. $g_3 = 0, a_3 = 0$: The line ... cannot be projected.
2. $g_3 = 0, a_3 \neq 0$: The line is **parallel to the image plane**, does not lie in the camera's xy -plane, and is projected to a parallel line $G'(\lambda)$ in the image plane

$$G'(\lambda) = -\frac{f}{a_3} \left(\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \lambda \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} \right).$$

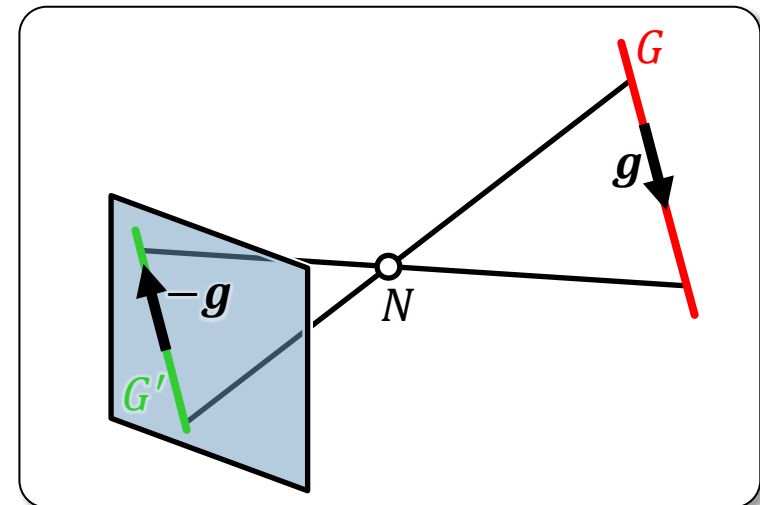
3. ...

4. ...

5. ...

Reminder

$$G'(\lambda) = -\frac{f}{a_3 + \lambda g_3} \left(\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \lambda \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} \right)$$



2.2 Perspective Projection

Perspective projection of lines (4)

1. $g_3 = 0, a_3 = 0$: The line ... cannot be projected.
2. $g_3 = 0, a_3 \neq 0$: The line ... is projected to a parallel line in image plane.
3. $g_3 \neq 0, a = 0$: The line passes through the center of projection N and is projected to

$$G'(\lambda) = -\frac{f}{\lambda g_3} \left(\lambda \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} \right) = -\frac{f}{g_3} \begin{pmatrix} g_1 \\ g_2 \end{pmatrix},$$

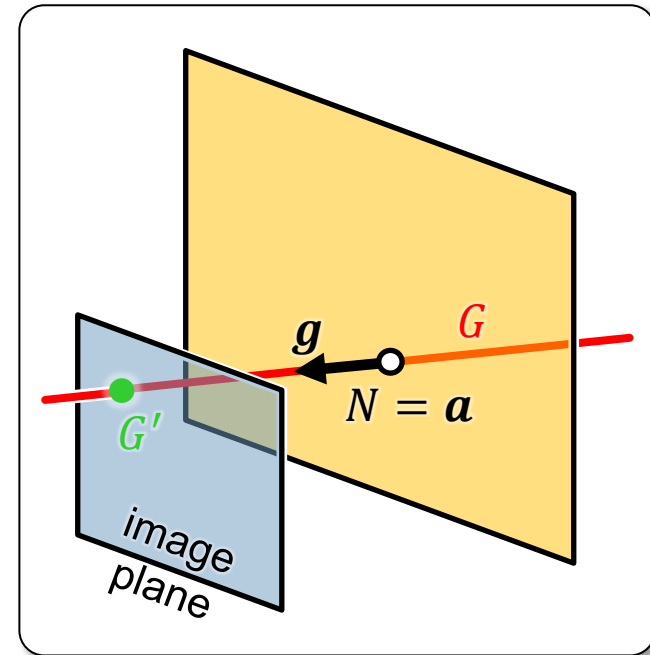
which is just a single point in the image plane.

4. ...

5. ...

Reminder

$$G'(\lambda) = -\frac{f}{a_3 + \lambda g_3} \left(\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \lambda \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} \right)$$



2.2 Perspective Projection

Perspective projection of lines (5)

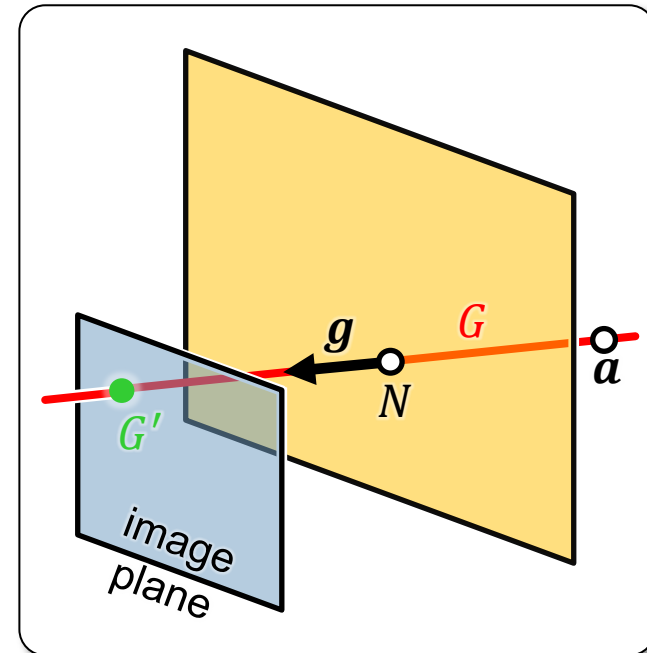
1. $g_3 = 0, a_3 = 0$: The line ... cannot be projected.
2. $g_3 = 0, a_3 \neq 0$: The line ... is projected to a parallel line in image plane.
3. $g_3 \neq 0, a = 0$: The line ... is projected to a single point in image plane.
4. $g_3 \neq 0, a = \lambda_0 g$: The line **passes through the center of projection** and is projected to a single point in the image plane

$$\begin{aligned} G'(\lambda) &= -\frac{f}{(\lambda + \lambda_0)g_3} \left((\lambda + \lambda_0) \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} \right) \\ &= -\frac{f}{g_3} \begin{pmatrix} g_1 \\ g_2 \end{pmatrix}. \end{aligned}$$

- This is a generalization of case 3.

Reminder

$$G'(\lambda) = -\frac{f}{a_3 + \lambda g_3} \left(\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \lambda \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} \right)$$



2.2 Perspective Projection

Perspective projection of lines (6)

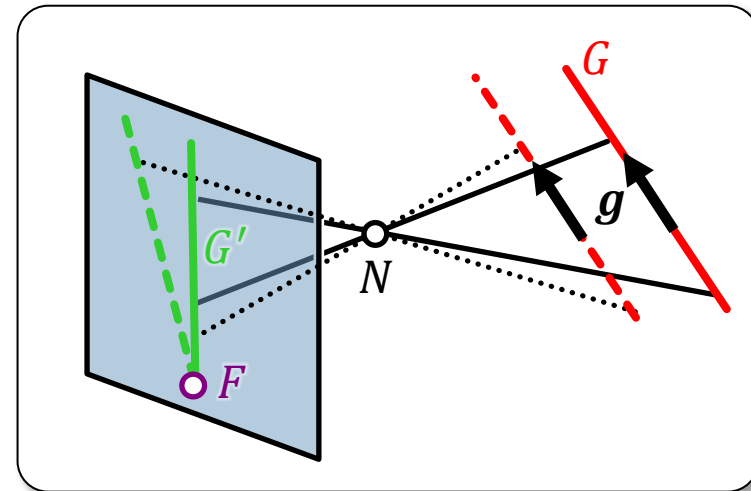
1. $g_3 = 0, a_3 = 0$: The line ... cannot be projected.
2. $g_3 = 0, a_3 \neq 0$: The line ... is projected to a parallel line in image plane.
3. $g_3 \neq 0, \mathbf{a} = 0$: The line ... is projected to a single point in image plane.
4. $g_3 \neq 0, \mathbf{a} = \lambda_0 \mathbf{g}$: The line ... is projected to a single point in image plane.
5. A line in **general position** is projected to a ray with **vanishing point F** (dt. Fluchtpunkt) in the image plane

$$\lim_{\lambda \rightarrow \infty} G'(\lambda) = -\frac{f}{g_3} \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} =: F.$$

- ▶ Parallel lines with identical $\mathbf{g}$ have the same vanishing point F .
- ▶ They intersect in their common vanishing point F in the image plane.

Reminder

$$G'(\lambda) = -\frac{f}{a_3 + \lambda g_3} \left(\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \lambda \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} \right)$$



2.2 Perspective Projection

Perspective projection of lines (7)

1.... 4.

5. A line in **general position** is projected to a ray with **vanishing point F** (dt. Fluchtpunkt) in the image plane

$$\lim_{\lambda \rightarrow \infty} G'(\lambda) = -\frac{f}{g_3} \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} =: F.$$

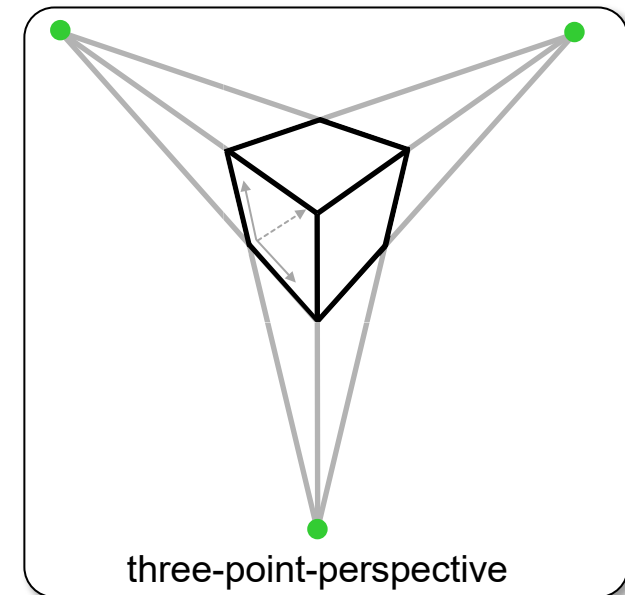
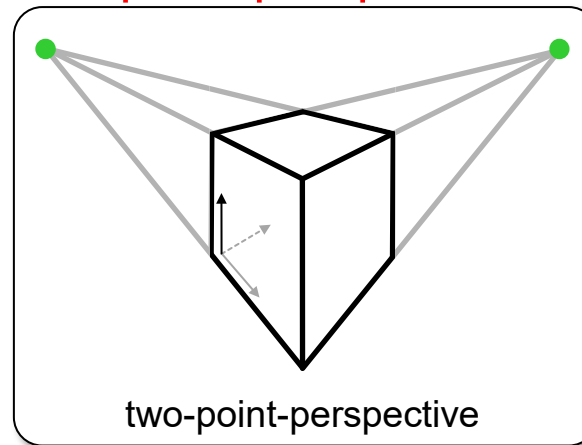
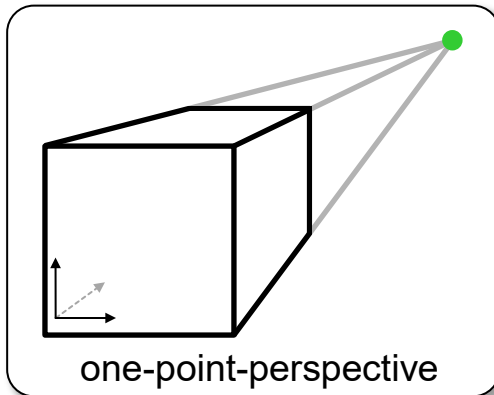
- ▶ Parallel lines with the same g have the same vanishing point F .
- ▶ They intersect in their common vanishing point F in the image plane.



2.2 Perspective Projection

Perspective projection of lines (8)

- ▶ In general, images of parallel lines intersect in a vanishing point.
- ▶ Every point in image plane is vanishing point of a bundle of parallel lines.
- ▶ If the coordinate axes of the world system are in general position, the coordinate axes correspond to **vanishing points**, if they are not parallel to the image plane.
- ▶ The respective perspective projections are called **one-**, **two-** or **three-point-perspectives**.

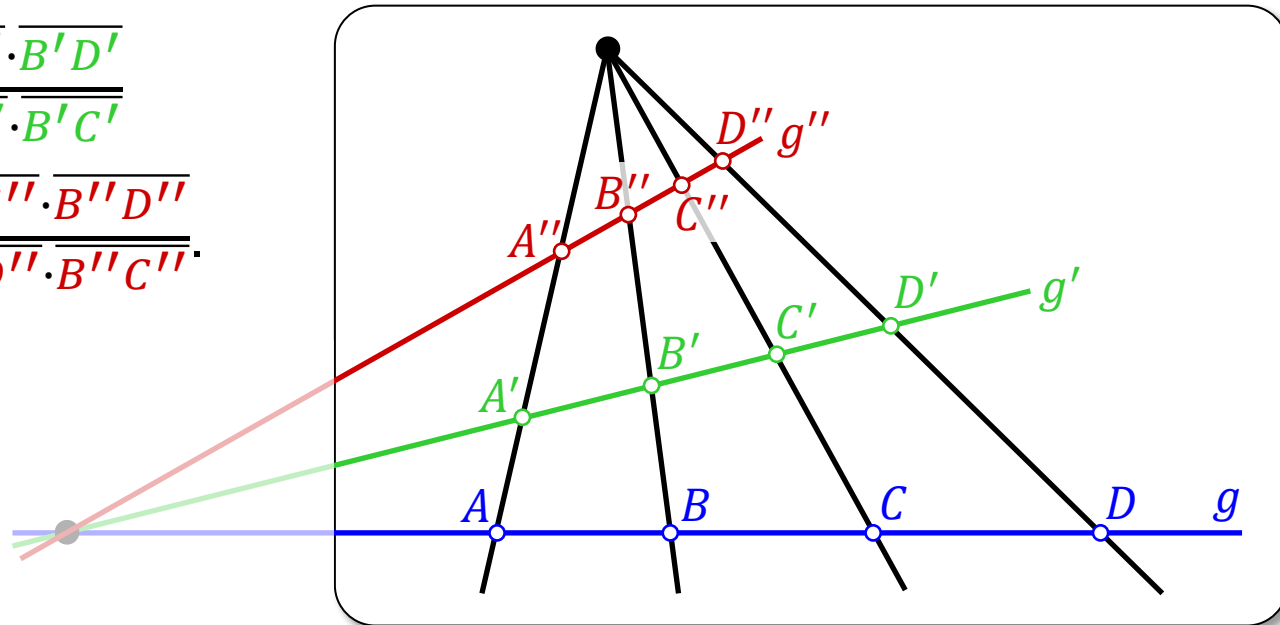


2.2 Perspective Projection

Perspective projection of lines (9)

- A line, not passing through the center of projection, is mapped to a half-line in the image plane.
- Length, parallel lines, angles, and ratios can change.
- **Cross-ratios** (dt.: Doppelverhältnis) are invariant:

$$\frac{\overline{AC} \cdot \overline{BD}}{\overline{AD} \cdot \overline{BC}} = \frac{\overline{A'C'} \cdot \overline{B'D'}}{\overline{A'D'} \cdot \overline{B'C'}} = \frac{\overline{A''C''} \cdot \overline{B''D''}}{\overline{A''D''} \cdot \overline{B''C''}}$$



2.2 Perspective Projection

General Perspective camera model

- So far, we used a **special setup**, where the camera system is the world system.

- **Extrinsic camera parameters**

dof = degrees of freedom

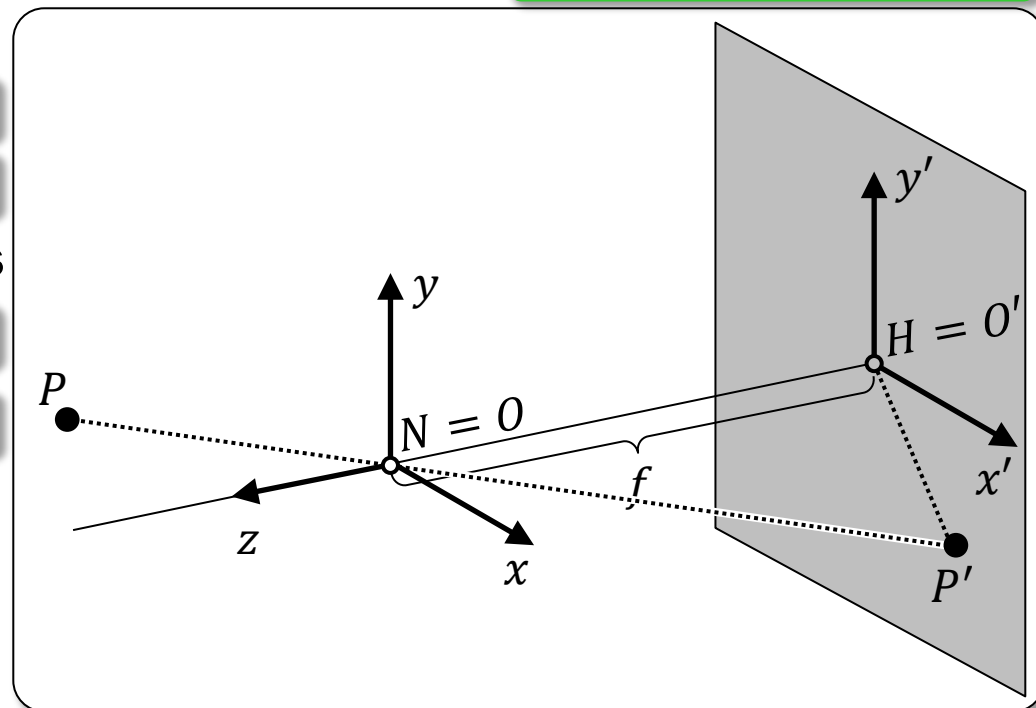
- Camera alignment as $M_b(\alpha)$ or Euler-angles. ← 3 dof
- Projection center N . ← 3 dof

- **Intrinsic camera parameters**

- Principal parameter f . ← 1 dof
- Principal point H . ← 2 dof

- **Calibration**

- Determine the nine camera parameters from images of reference objects.



2.2 Perspective Projection

General perspective projection

- Translation of the center of projection N , i.e., $N \neq O$.
 - Rotation of the camera system by $M_b(\alpha) \in SO(3) \subseteq \mathbb{R}^{3 \times 3}$.
 - Translation of image principal point H , i.e., $H \neq O'$.
- Similarity transformation (with $s = 1$, aka isometry, rigid body motion) of the camera system.

2.2 Perspective Projection

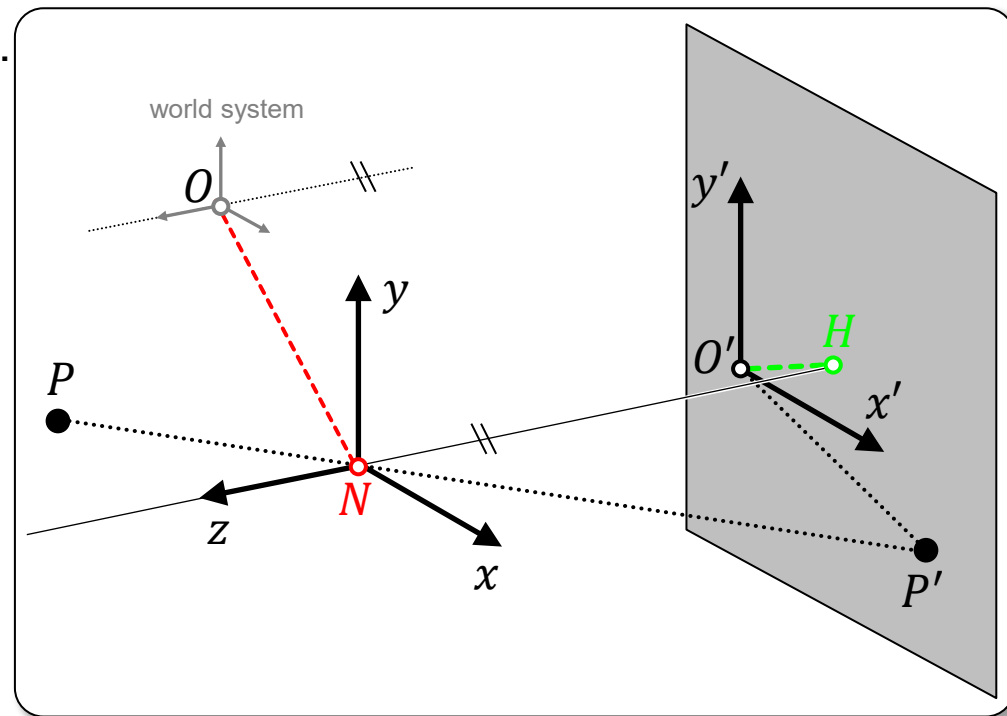
■ Perspective projection with translation of the

- center of projection $N = \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix}$ and
- image principal point $H = \begin{pmatrix} x'_0 \\ y'_0 \end{pmatrix}$.

► Mapping (affine version):

$$\begin{pmatrix} x' - x'_0 \\ y' - y'_0 \end{pmatrix} = -\frac{f}{z - z_0} \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix}$$

for $z - z_0 \neq 0$.



2.2 Perspective Projection

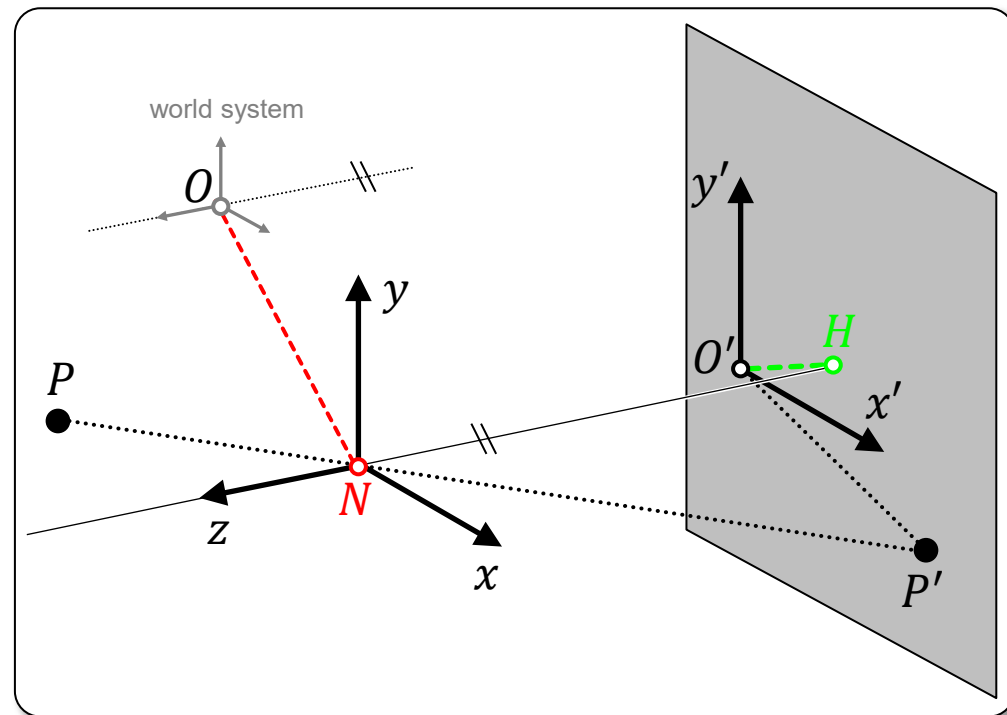
► In homogeneous coordinates

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} \sim \begin{pmatrix} f & 0 & x'_0 \\ 0 & f & y'_0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & -x_0 \\ 0 & 1 & 0 & -y_0 \\ 0 & 0 & 1 & -z_0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

► Mapping (affine version):

$$\begin{pmatrix} x' - x'_0 \\ y' - y'_0 \end{pmatrix} = -\frac{f}{z - z_0} \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix}$$

for $z - z_0 \neq 0$.



2.2 Perspective Projection

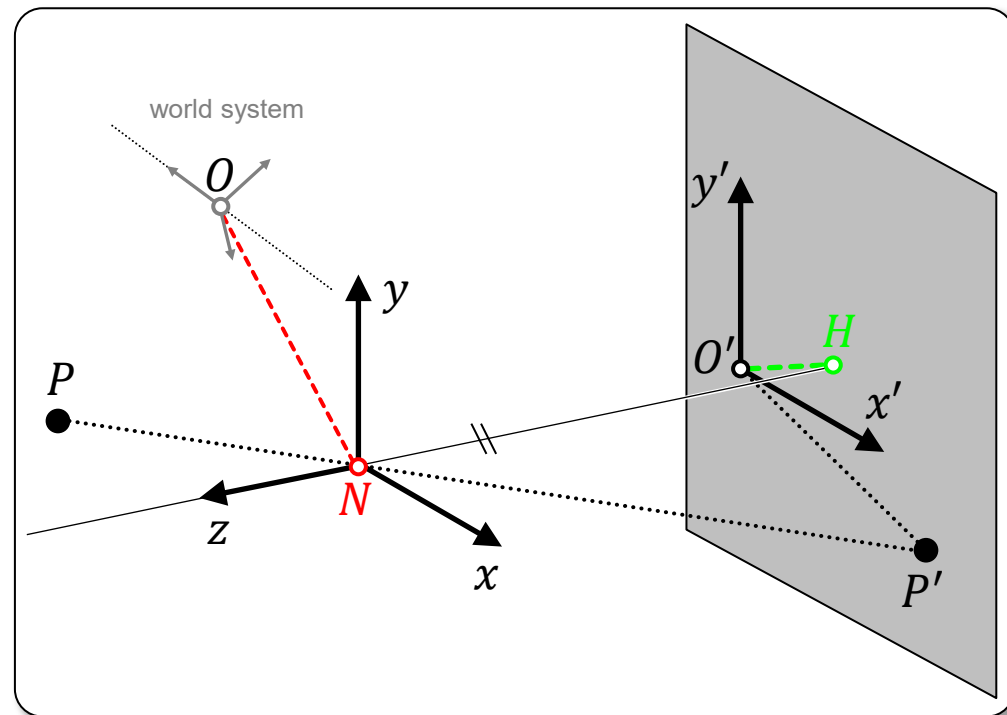
- Perspective projection with translation of N and H and
 - rotation of the camera system with $R = (r_{ij}) = M_b(\alpha) \in SO(3)$,
i.e., $R^t \cdot N = (\tilde{x}_0, \tilde{y}_0, \tilde{z}_0)^t$

► Mapping (affine version):

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x'_0 \\ y'_0 \end{pmatrix} - \frac{f}{z_c} \begin{pmatrix} x_c \\ y_c \end{pmatrix},$$

with $z_c \neq 0$ and

$$\begin{pmatrix} x_c \\ y_c \\ z_c \end{pmatrix} = R^t \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - N \right).$$



2.2 Perspective Projection

► In homogeneous coordinates

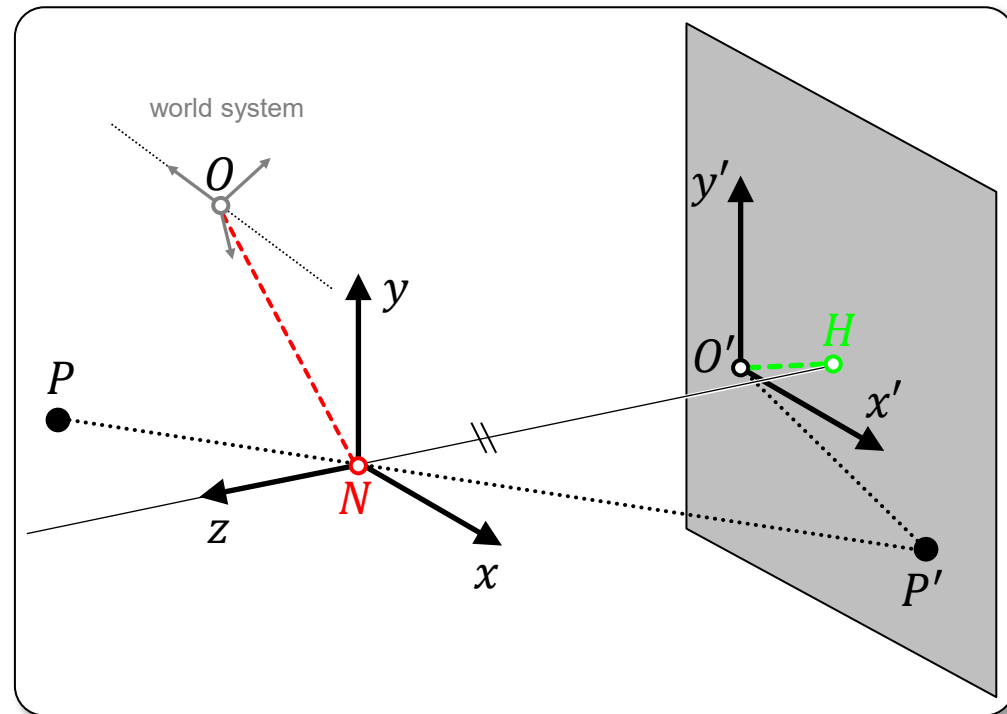
$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} \sim \begin{pmatrix} f & 0 & x'_0 \\ 0 & f & y'_0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} r_{11} & r_{21} & r_{31} & -\tilde{x}_0 \\ r_{12} & r_{22} & r_{32} & -\tilde{y}_0 \\ r_{13} & r_{23} & r_{33} & -\tilde{z}_0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} f \cdot I & H \\ 0 & -1 \end{pmatrix} (R^t, -R^t N) \begin{pmatrix} P \\ 1 \end{pmatrix}$$

► Mapping (affine version):

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x'_0 \\ y'_0 \end{pmatrix} - \frac{f}{z_c} \begin{pmatrix} x_c \\ y_c \end{pmatrix},$$

with $z_c \neq 0$ and

$$\begin{pmatrix} x_c \\ y_c \\ z_c \end{pmatrix} = R^t \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - N \right).$$



2.2 Perspective Projection

...for $z_c \neq 0$.

► In homogeneous coordinates

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} \sim \underbrace{\begin{pmatrix} f_x & 0 & x'_0 \\ 0 & f_y & y'_0 \\ 0 & 0 & -1 \end{pmatrix}}_{= K \in \mathbb{R}^{3 \times 3}} \cdot \underbrace{\begin{pmatrix} r_{11} & r_{21} & r_{31} & -\tilde{x}_0 \\ r_{12} & r_{22} & r_{32} & -\tilde{y}_0 \\ r_{13} & r_{23} & r_{33} & -\tilde{z}_0 \end{pmatrix}}_{= (R^t | -R^t N) \in \mathbb{R}^{3 \times 4}} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} \quad \text{B}$$

intrinsic camera parameters, aka **camera calibration matrix** extrinsic camera parameters

all camera parameters, aka **camera matrix**

► Each object point $(x, y, z)^t$ yields exactly one image point $(x', y')^t$

$$\begin{aligned} x' &= x'_0 - f \frac{r_{11}(x-x_0)+r_{21}(y-y_0)+r_{31}(z-z_0)}{r_{13}(x-x_0)+r_{23}(y-y_0)+r_{33}(z-z_0)}, \\ y' &= y'_0 - f \frac{r_{12}(x-x_0)+r_{22}(y-y_0)+r_{32}(z-z_0)}{r_{13}(x-x_0)+r_{23}(y-y_0)+r_{33}(z-z_0)}. \end{aligned} \quad \text{C}$$

2.2 Perspective Projection

- General perspective projection with
 - translation of the center of projection $N = (x_0, y_0, z_0)^t$,
 - translation of the image principal point $H = (x'_0, y'_0)^t$, and
 - rotation of the camera system with $R = (r_{ij}) = M_b(\alpha) \in SO(3)$,
i.e., $R^t \cdot N = (\tilde{x}_0, \tilde{y}_0, \tilde{z}_0)^t$
- Three equivalent versions for the general perspective projection:
 - A affine version
 - B homogeneous version
 - C explicit version

2.2 Perspective Projection

Spatial reconstruction from perspective projections (1)

- Each object point $(x, y, z)^t$ yields exactly one image point $(x', y')^t$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x'_0 \\ y'_0 \end{pmatrix} - \frac{f}{z_c} \begin{pmatrix} x_c \\ y_c \end{pmatrix} \text{ with } \begin{pmatrix} x_c \\ y_c \\ z_c \end{pmatrix} = R^t \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - N \right), \text{ if } z_c \neq 0.$$

- Solve these equations for the object coordinates x, y, z

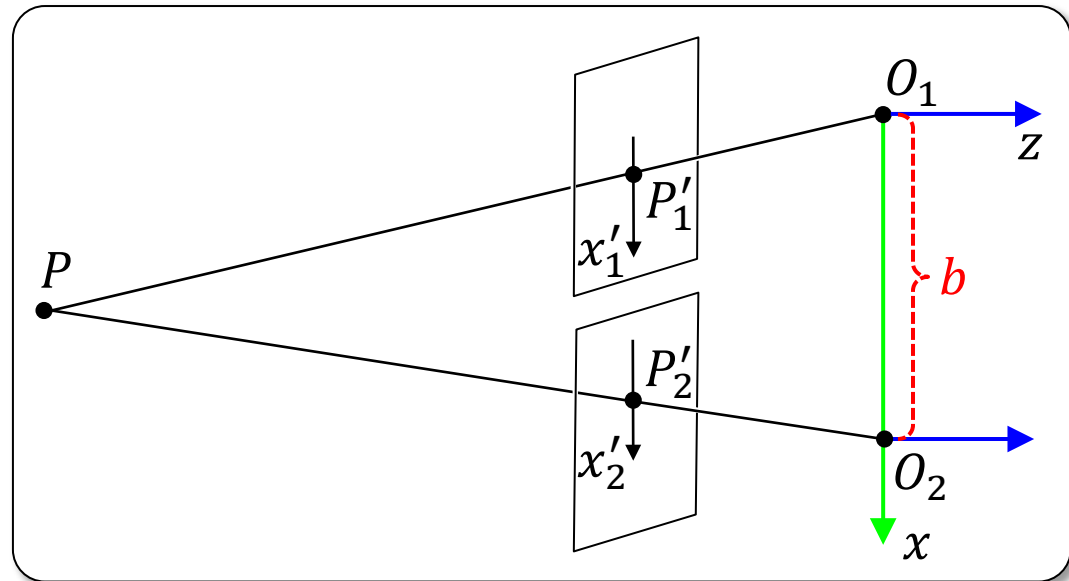
$$\left. \begin{aligned} x &= x_0 + (z - z_0) \frac{r_{11}(x' - x'_0) + r_{12}(y' - y'_0) - r_{13}f}{r_{31}(x' - x'_0) + r_{32}(y' - y'_0) - r_{33}f}, \\ y &= y_0 + (z - z_0) \frac{r_{21}(x' - x'_0) + r_{22}(y' - y'_0) - r_{23}f}{r_{31}(x' - x'_0) + r_{32}(y' - y'_0) - r_{33}f}. \end{aligned} \right\} (*)$$

- Each image point $(x', y')^t$ corresponds to infinitely many object points.
- A single image does not suffice to reconstruct a spatial object.

2.2 Perspective Projection

Spatial reconstruction from perspective projections (2)

- For the spatial reconstruction, several **correspondences** are required (provided the camera parameters are known), i.e., 3d object points, that can be seen in multiple 2d images from different perspectives.
- ▶ One object point $P = (x, y, z)^t$ visible in two different image points $P'_1 = (x'_1, y'_1)^t$ and $P'_2 = (x'_2, y'_2)^t$, yields
 - four equations (twice $(\star)$) containing
 - the three unknowns x, y, z .
- ▶ **First**, a good calibration is required to determine all camera parameters.



2.2 Perspective Projection

Calibration (1)

- Methods to determine the unknown camera parameters are called **calibration**.
- It typically uses **reference points** (aka ground truth points) on a calibration object
 - Points, for which the image **and** world coordinates are known.

► Calibration equations:

$$x = x_0 + (z - z_0) \frac{r_{11}(x' - x'_0) + r_{12}(y' - y'_0) - r_{13}f}{r_{31}(x' - x'_0) + r_{32}(y' - y'_0) - r_{33}f},$$
$$y = y_0 + (z - z_0) \frac{r_{21}(x' - x'_0) + r_{22}(y' - y'_0) - r_{23}f}{r_{31}(x' - x'_0) + r_{32}(y' - y'_0) - r_{33}f}.$$

- Each reference point yields two equations.
- To determine all nine camera parameters, at least five reference points are necessary.
 - If only the rotation is of interest, two reference points suffice.

2.2 Perspective Projection

Calibration (2)

Some problems are...

- noisy measurements,
 - wrong correspondences,
 - physics of real cameras, e.g.
 - low light intensity of pinhole cameras
 - ▶ use lenses to accumulate light.
 - ▶ lens distortion (German: Verzeichnung), etc.
 - refraction at transitions of materials with different refraction index,
 - dependent on wavelength, i.e., chromatic aberration.
 - the image plane is rasterized in pixels.
 - ...
- ▶ Choose the reference points (calibration object) as simple as possible.

Wikipedia



2.2 Perspective Projection

Central projection for planar objects (1)

- Choose the world coordinate system, such that a planar object is in the xy -plane, i.e., $z = 0$.
- The calibration equations

$$x = x_0 + (z - z_0) \frac{r_{11}(x' - x'_0) + r_{12}(y' - y'_0) - r_{13}f}{r_{31}(x' - x'_0) + r_{32}(y' - y'_0) - r_{33}f},$$

$$y = y_0 + (z - z_0) \frac{r_{21}(x' - x'_0) + r_{22}(y' - y'_0) - r_{23}f}{r_{31}(x' - x'_0) + r_{32}(y' - y'_0) - r_{33}f}.$$

simplify to

$$x = \frac{a_1 x' + a_2 y' + a_3}{c_1 x' + c_2 y' + 1},$$

$$y = \frac{b_1 x' + b_2 y' + b_3}{c_1 x' + c_2 y' + 1}, \quad \text{with, e.g., } a_1 = \frac{z_0 r_{11} - x_0 r_{31}}{r_{33} f}, \quad a_2 = \dots$$

2.2 Perspective Projection

Central projection for planar objects (2)

► Reconstruction:

One image suffices for the reconstruction of such a planar calibration object.

► Calibration:

The central projection of a planar object is determined by eight parameters.

► This requires $n \geq 4$ reference points.

► Each reference point $(x_i, y_i, 0)^t, (x'_i, y'_i)^t, i = 1, \dots, n$, yields two linear equations (multiplication with denominator):

$$x_i = x'_i a_1 + y'_i a_2 + a_3 - x_i x'_i c_1 - x_i y'_i c_2,$$

$$y_i = x'_i b_1 + y'_i b_2 + b_3 - y_i x'_i c_1 - y_i y'_i c_2.$$

► ... or in a $n \times 8$ design matrix M ...

2.2 Perspective Projection

- ▶ ... or in a $n \times 8$ design matrix M ...

$$\begin{pmatrix} x'_1 & y'_1 & 1 & 0 & 0 & 0 & -x_1x'_1 & -x_1y'_1 \\ 0 & 0 & 0 & x'_1 & y'_1 & 1 & -y_1x'_1 & -y_1y'_1 \\ x'_2 & y'_2 & 1 & 0 & 0 & 0 & -x_2x'_2 & -x_2y'_2 \\ 0 & 0 & 0 & x'_2 & y'_2 & 1 & -y_2x'_2 & -y_2y'_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x'_n & y'_n & 1 & 0 & 0 & 0 & -x_nx'_n & -x_ny'_n \\ 0 & 0 & 0 & x'_n & y'_n & 1 & -y_nx'_n & -y_ny'_n \end{pmatrix} \cdot \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ b_1 \\ b_2 \\ b_3 \\ c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \\ x_2 \\ y_2 \\ \vdots \\ x_n \\ y_n \end{pmatrix}$$

... for short: $M \cdot \mathbf{a} = \mathbf{x}$.

- ▶ The unknown parameters are given by $\mathbf{a} = M^{-1}\mathbf{x}$.
- ▶ Because M is over-determined, there is in general no inverse matrix.
 - ▶ Use least squares method ...
 - ▶ ... use singular value decomposition (SVD)!!

Content

- 1. Affine Geometry ✓
 - 1.1 2d Affine Geometry ✓
 - 1.2 3d Affine Geometry ✓
- 2. Projective Geometry ✓
 - 2.1 Parallel Projection ✓
 - 2.2 Perspective Projection ✓
- 3. Physics of Camera Projections*

3. Physics of Camera Projections*

Optics

- Optics is an area in physics, dealing with the properties of light and its interaction with other physical materials.
- There are various models to describe different phenomena of light, e.g.
 - light rays, aka geometric optics,
 - electromagnetic waves/radiation, aka wave or physical optics,
 - electromagnetic energy transport, aka photometry or radiometry,
 - photons, aka quantum electrodynamics.
- In **geometric optics**, light is modelled as light rays to approximate the propagation of light.
- In computer vision, often only **paraxial optics** is used, where the angles of the rays to the optical axis are relatively small.

3. Physics of Camera Projections*

Reflection and refraction

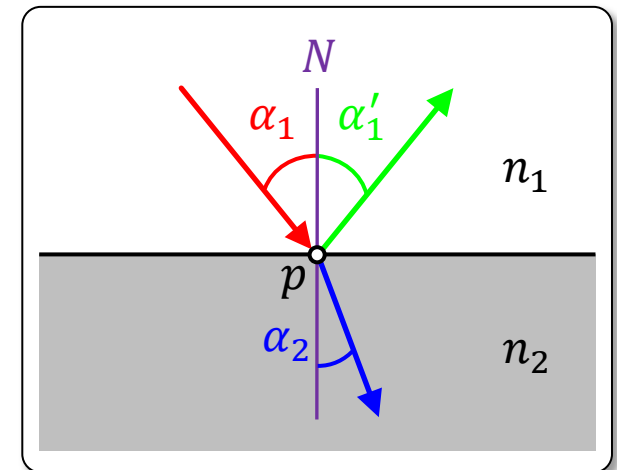
- **Reflection** happens at the boundary surface of an opaque material, where the **incoming light ray** is reflected at the intersection point p and **continues** at angle $\alpha_1 = \alpha'_1$ to the **surface normal** N .
- **Refraction** happens at the boundary surface of a transparent or translucent object, where the **incoming light ray** enters the object and **continues** at angle α_2 to the **surface normal** N with

$$n_1 \sin \alpha_1 = n_2 \sin \alpha_2,$$

where $n_1 < n_2$ are the refraction indices.

In paraxial optics: $n_1 \alpha_1 = n_2 \alpha_2$.

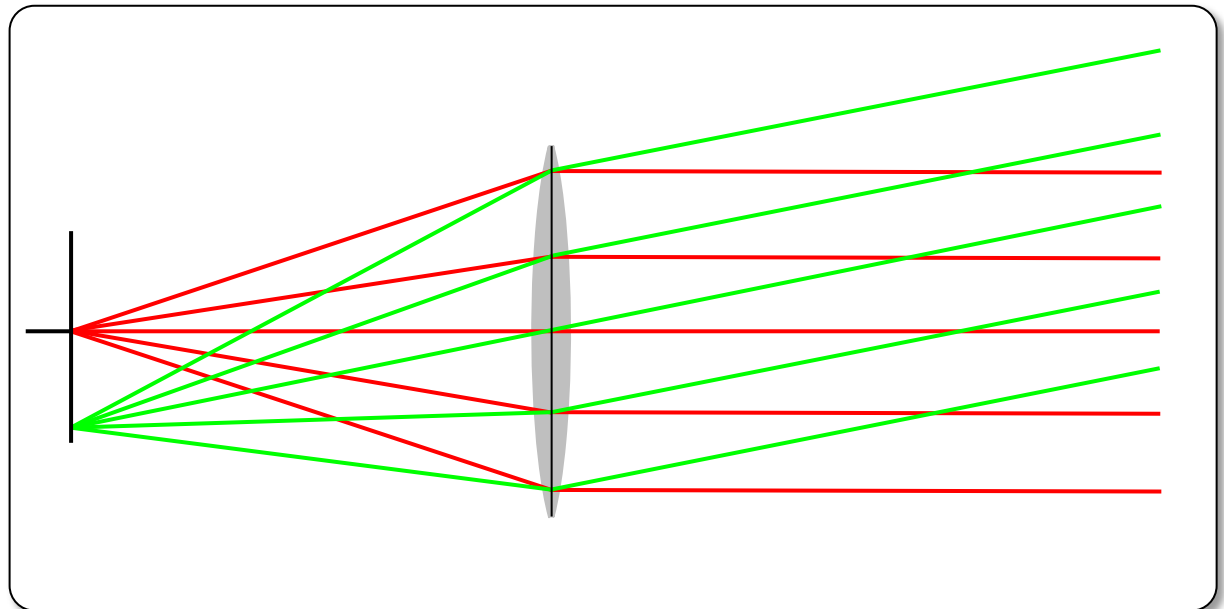
- **Important note:** Both phenomena are planar, i.e., they happen in the plane spanned by the **incoming light ray** and the **surface normal** N .



3. Physics of Camera Projections*

Light Intensity

- Pinhole cameras yield low light intensities, because the light in a single point on the image plane comes from a very small cone in space.
- In practice, lenses are used, where parallel rays or multiple rays from individual object points are mapped onto a single image point.
- ▶ The light intensity in the image point is increased.

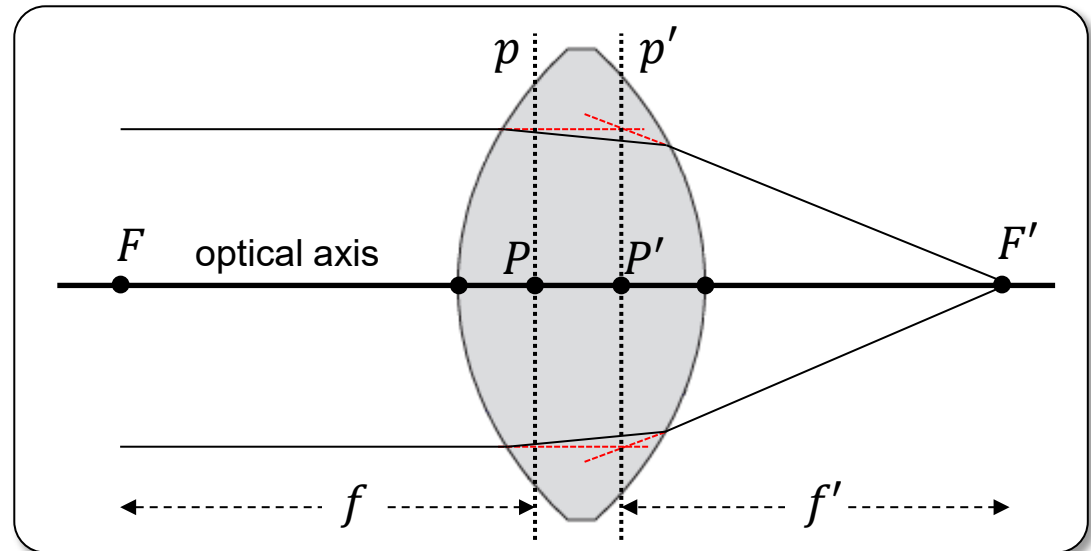


[according to Geissler 1999]

3. Physics of Camera Projections*

Properties of Lenses (1)

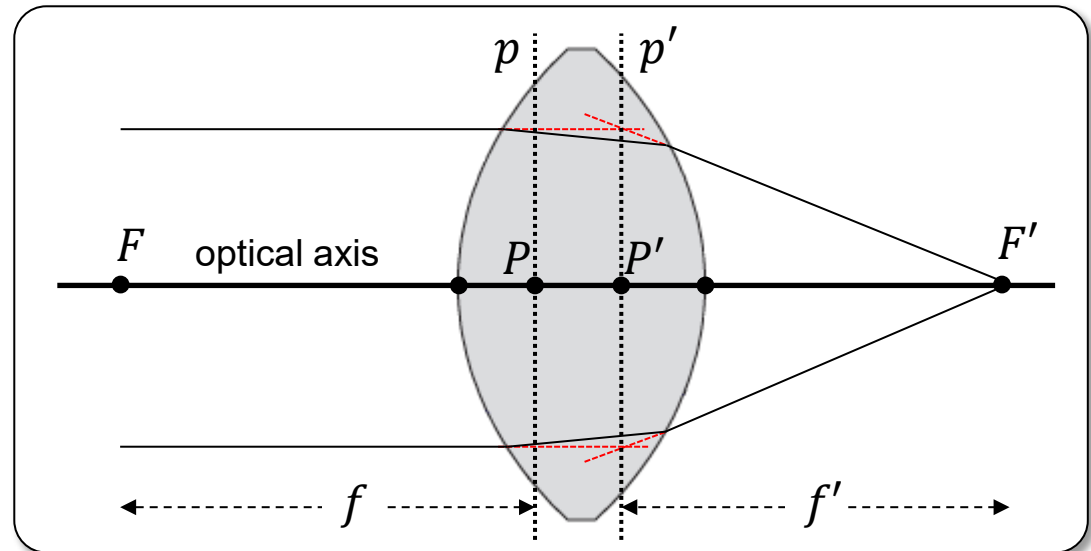
- **Optical axis:** Symmetry axis of a lens.
- **Focal point:** All rays parallel to the optical axis intersect in the **focal point** F' on the image side of the lens.
- **Principal planes:** Incoming and outgoing rays intersect in lens **principal planes** (one on the object side p and one on the image side p').
- **Principal points:** Intersections P, P' of principal planes with optical axis.
- **Focal distance:** The distance between image-side principal point and the image-side focal point is called **focal distance** f' .



3. Physics of Camera Projections*

Properties of Lenses (2)

- **Construction of an optical path:** A ray from the object, parallel to the optical axis, is *formally* refracted at the lens' image-side principal plane, such that it passes the focal point F' .
- **Remark:** This is only an approximation to the real, physically correct optical path.
 - It is easy to compute.
- **Chromatic aberration:** The refraction index depends on the wavelength.
 - ▶ The focal distance depends also on the wavelength.



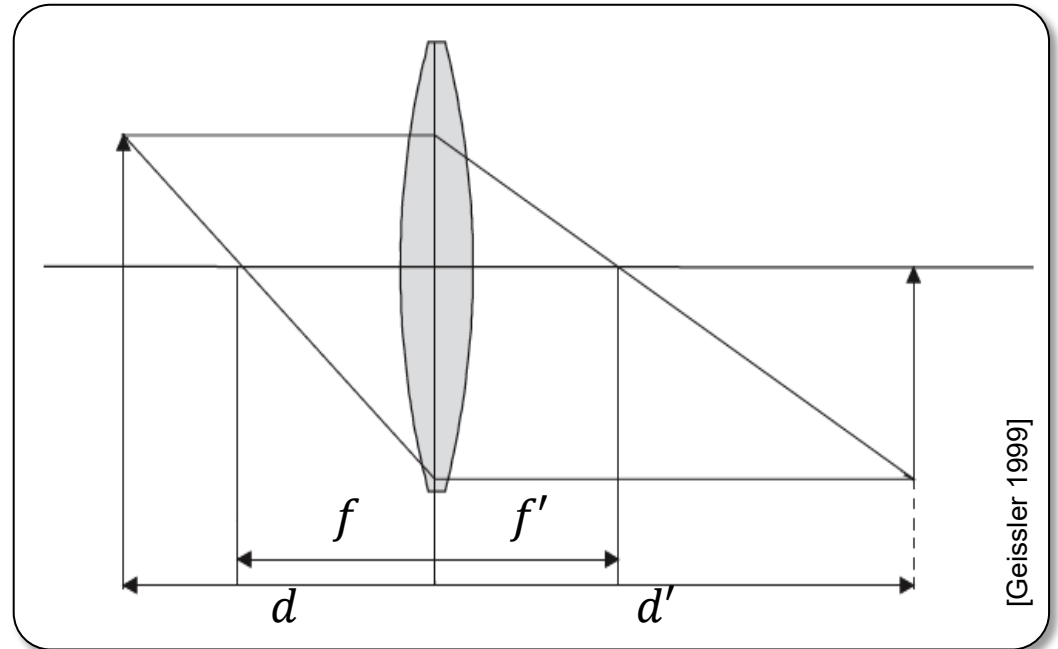
[Geissler 1999]

3. Physics of Camera Projections*

Thin Lenses (1)

- For thin lenses, the principal planes approximately coalesce.
- In this case, the lens equation simplifies to (d : object distance, d' : image distance):

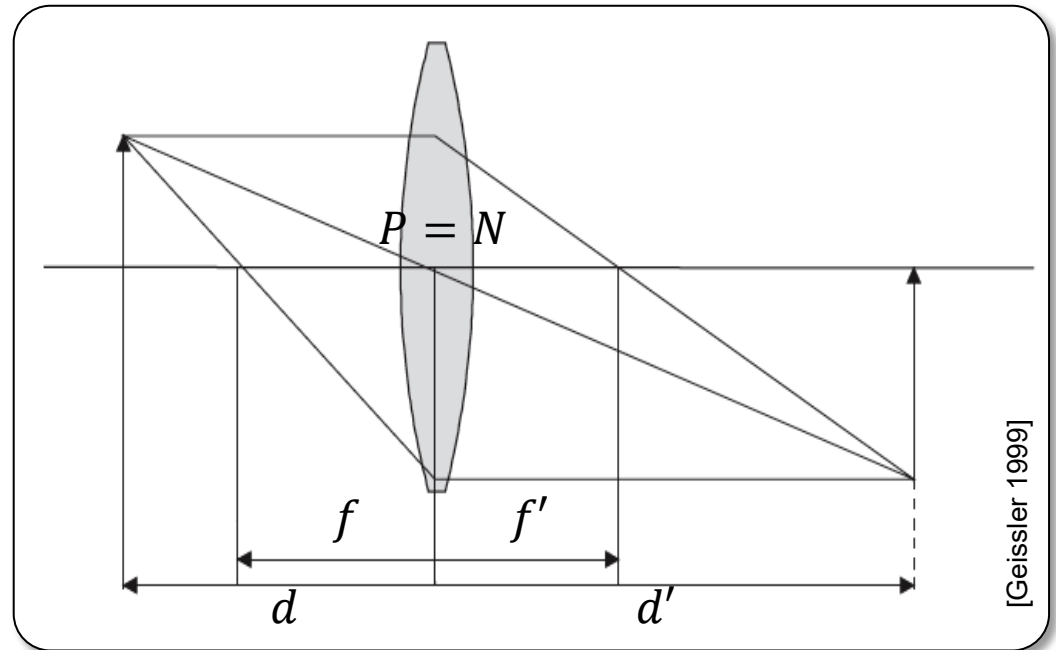
$$\frac{1}{d} + \frac{1}{d'} = \frac{1}{f'}.$$



3. Physics of Camera Projections*

Thin Lenses (2)

- **Knot point:** Rays through the principal point pass the lens without refraction.
- ▶ The lens principal point is equal to the knot point.



- **Disadvantage compared to a pin hole camera:** For different object distances the image distance (or the refraction index of the lens) must be accommodated.
- ▶ The field of depth is limited.

Questions

- What is a Cartesian coordinate system?
- How are translations, rotations, and scalings represented in 2d and 3d?
- What is an affine transformation?
- What is a rigid body motion?
- What are homogenous coordinates?
- What is a perspective projection?
- What is a parallel projection?
- What are the parameters in a perspective projection camera system?
- How do you compute a perspective projection?
- What is the problem in a spatial reconstruction from perspective projections?
- What are reference points?
- What is camera calibration?